MAgent Theory 2.0

Formation with Top-Down Feedback:
Shared Labels and Exact Parents

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Status and summary

This working document is not part of the main manuscript and is not cited by it. It asks whether parents that are created by a state-determined handoff when a wall forms, and that then act back on their members, give an inherited recursive formation in the sense of the formation standard of the main manuscript, stated in its chapter *Inherited Parents and Recursive Formation*, which we call the inheritance chapter. The clauses (F1) to (F6) of that standard, its two null flows and its frozen-environment control are restated in definition 6.1, so that the verdicts below can be read without the main manuscript. The constructions are stated for general probability laws. The Gaussian location family is the computational instance, and every statement that depends on it says so.

Feedback of the obvious kinds does not change what forms. A parent added as one more source in its members' rows multiplies the row part of each member's force by a common factor when the member agrees with it, leaves record forces unchanged, and alters no exponent of the rows (proposition 2.1); in an example without records the same blocks form and only the second stage is delayed. In an objective with a gradient flow, such as the global joint of Items 2 and 3 or the reciprocal potential with fixed rows, a top-down cost invariant under the collective motion of a block exerts no force on the block's momentum whenever that motion has a momentum function, as it does in the Gaussian location family (proposition 2.2). An exactly inherited pooled parent therefore does not see it; the sharpening of walls reported in Item 3 passes through the distribution of that momentum among the members, which sparse bridges read. A post-handoff flow with a globally attracting equilibrium cannot meet the counterfactual clause (F6b) when persistence means that the candidate relation holds at all later times, the equilibrium lies off the boundary of the candidate set, and no competing block is selected first (proposition 2.3). The selector used here selects after a finite dwell, and for such selectors a unique attractor excludes nothing: a strictly convex Gaussian hierarchy in a constant metric can meet (F6b), because the length of a transient dwell in the candidate set depends on the handoff (remark 2.4). In the Gaussian tangent global joint of Items 2 and 3, run in its own Fisher metric, such instances were found numerically only with declared parent anchors that pull the parents together.

What changes outcomes is the pooling of evidence. When a block forms, its members share one source label: they infer jointly which source to trust, with the prior over sources applied once and their costs added. For general laws the shared row depends on the block only through the linear pool of the members' laws, and it equals the Gibbs row of a single receiver holding that pool at temperature T/n (proposition 3.2); for sources in an exponential family it depends only on the pooled mean parameter. The parent presents the pool to its members and to every receiver outside the block. With parents that present the pool, a tempered label, which averages the members' costs instead of adding them, gives the same vector field as the dynamics without feedback at every internally synchronized configuration (proposition 3.4), so the factor n in the exponent is the operative part of the feedback.

In the Gaussian location family the shared-label parent is exactly inherited between handoffs and compresses. The pooled mean of every block evolves autonomously, by the per-agent law of a single agent at temperature T/n with one additional source at its own law, while the members' deviations from the pool relax at a fixed rate to offsets determined by their records, whatever the spread within the block (theorem 4.1). For linear consensus on a fixed graph, exact block-mean dynamics over external equitable partitions were analyzed by Martini et al. [13] and Egerstedt et al. [4], and O'Clery et al. [17] relate them to observability and coarse graining; here the rows are state-dependent Gibbs rows. Universal pooled closure does not extend beyond Gaussian families of the sufficient statistic: if the pooled parent of a two-member block closes for every configuration and every target, the covariance of the sufficient statistic is constant and its laws are Gaussian with one covariance (theorem 4.5). In the coordinates of the sufficient statistic such a family is a Gaussian location family; the common-variance lognormal family on $(0, \infty)$ is an example whose laws are not Gaussian. For Bernoulli laws pooled closure fails.

In a twelve-agent instance with four groups of three conflicting records, the label-preserving null flow never selects a block, and the renormalized null selects only the block of all twelve agents (propositions 5.1 and 5.2). The shared-label flow selects the four groups and then, on each side, the union of two first-stage parents. The second stage reduces to a one-dimensional equation for the gap between two parents, with a stable merged state, an unstable middle root, and a stable separated state held by the records; interval arithmetic certifies that these are its only zeros (proposition 5.4). The union forms from every post-handoff state in the stated ranges whose gap lies below the middle root by more than 2×10^{-8}, and never from one whose gap lies above it by more than that (corollary 5.5). Post-handoff states shifted outside the middle root therefore never unite, while the control without feedback unites from every tested shift; these shifted states are not, however, the counterfactual class of the standard. The bistability exists only for record gaps above a saddle-node of the gap equation, certified to lie in $(1.0722, 1.0723)$. At slightly larger record gaps than the instance's the hierarchy stops at depth one in the separated state, and the distance from which two blocks still unite falls with block size, from 0.963 for blocks of three to 0.593 for blocks of six.

Measured against the standard, the instance is not shown to be a recursive formation, and its classification is undetermined (chapter 6). The admitted class is open in the full grade, so (F1) holds. The selector is declared with single values of its constants; it meets (F2) in form, but the standard asks for a declared set of constants with nonempty interior, over which (F5) is not established. Inheritance (F3) is proved for the dynamics and for read-only parent queries at a fixed partition; the process with candidate clocks, blocks of unequal size and merges at handoffs is not shown to descend from the retained state, so (F3) is open. Compression (F4) has numerical witnesses on the reachable set at the first stage; at the second stage it depends on whether the parent type records the time of its formation, which is not declared, so it is open there. Interaction necessity (F5) at the declared constants holds against both nulls on an open neighborhood of a tested state: both null histories are proved on the whole admitted class, while the history of the shared-label flow rests on fixed-step integration of the first stage with explicit margins, everything after the first-stage handoff following from certified inequalities. The frozen-environment control selects the four groups and no union. (F6a) holds on the tested states under the same proviso. (F6b) holds for the shifted post-handoff states by certified inequalities, but against the counterfactual class of the standard

it is open: no such class is declared, and widened initial states along which the first stage has not occurred by the actual handoff times give handoff gaps below the middle root, from which the union forms. Exactness holds only in Gaussian families of the sufficient statistic, the compressed coordinates are transient, the selector thresholds and the label-sharing handoff are declared, and the parameter window is narrow. A comparison with the literature found exact quotients of linear consensus, Bayesian inference of latent social groups, representative agents for coherent groups, and Bayesian structure learning of hierarchies, but no construction combining shared-label pooling, exact block-mean closure away from synchrony, and merging that depends on the state at formation (chapter 8). This is not a priority claim.

A derived selector replaces the declared thresholds by a comparison of free energies (chapter 7). A block is selected when the shared-label models that the handoff would create, together with the models of all other units, have a lower free energy than the current ones, and the comparison has held for the declared window. For general laws this structure free energy reads only the spread of each block, n times a generalized Jensen–Shannon divergence, and n times the free energy of its pool at temperature T/n, which includes the block's own parent at cost zero (proposition 7.2). Two agents whose mutual attentions have a product above the peer prior gain from a shared label, through a saving of labels and a double use of their laws in the pool, and the gain grows with their divergence while they attend to each other (proposition 7.3). The criterion detects attention, not agreement, and only blocks, not single agents, have a merge radius. The derived selector therefore groups whatever attention the initial state contains, and in the instance the label-preserving null selects the four groups at once. The criterion for uniting two parents is a radius, $d_* = 0.7685$ for blocks of three, below the middle root of the gap equation (proposition 7.6), so the dependence on the post-handoff gap survives (corollary 7.7). With $\tau_p = 1.25$ the label-preserving null and the frozen-environment control select the four groups and nothing more, the renormalized null selects the two sides and then all agents, and the shared-label flow selects the four groups and then their unions. The window decides the regime: with an earlier handoff the hierarchy stops at depth one, as the label-preserving null does, a handoff in a short later interval gives a mixed regime, and with the window $\tau_p = 2$ of the declared instance the flow selects the two unions directly as a single stage.

Every identity was checked numerically at small sizes in the accompanying checks, numbered (1) to (28) in the text; the checks support the instances tested and are not proofs. Check (27) is the exception: it certifies the zeros and sign margins of the reduced gap equations and the union radius by outward-rounded interval arithmetic, so those inequalities are proved. Statements that rest on numerical integration are labeled as such.

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Chapter 1

Setting

The agents $i \in \mathcal{V}$, $|\mathcal{V}| = N$, hold belief laws q_i on a standard Borel space Y. The model sector is trivial in this document; it enters only through a common evaluator G, a law on Y shared by agents of one species. In the per-agent mixture-source model of the dynamics chapter of the main manuscript (*Free Energies and Coupled Dynamics of Agent Laws*), agent i has a finite source set $\mathcal{N}_i$ containing its own label, with positive prior row π_i. Its self cost is $D_{ii} = D_{\mathrm{KL}}(q_i \| G)$, its cost for a peer j is $D_{ij} = D_{\mathrm{KL}}(q_i \| \nu_{ij})$ with ν_{ij} the law of j as compared by i, and its records o_i enter through a likelihood $\ell_i(o_i \mid k)$ that persists in the objective. At temperature T the per-agent free energy is

$$\mathcal{F}_i = \sum_{\nu \in \mathcal{N}_i} \beta_{i\nu} D_{i\nu} + T D_{\mathrm{KL}}(\beta_i \| \pi_i) + \int -\log \ell_i(o_i \mid k) q_i(dk), \tag{1.1}$$

and its minimizing row is the Gibbs row $\beta_{i\nu} = \pi_{i\nu} e^{-D_{i\nu}/T} / Z_i$. The main manuscript's per-agent model has $T = 1$; another temperature corresponds to tempered factors. Each agent follows the Fisher gradient flow of its own $\mathcal{F}_i$, with the laws of the other agents held as data. By the envelope identity, the differential of $\mathcal{F}_i$ at its Gibbs row is $\sum_\nu \beta_{i\nu} dD_{i\nu}$ plus the differential of the record term, so the force on q_i is a row-weighted sum of divergence forces plus a record force that no row weights.

Definition 1.1 (Units, presented laws and the wall selector). At each time the agents are partitioned into units: the blocks selected so far that are contained in no later block, and the agents that belong to no block. Each unit U presents a law P_U, which is the agent's own law for an agent and is declared by the handoff of definition 3.1 for a block. Fix thresholds $0 < \delta_c < \delta_w$ and a window $\tau_p > 0$. Units $U \neq V$ are coherent if $D_{\mathrm{KL}}(P_U \| P_V) < \delta_c$ and $D_{\mathrm{KL}}(P_V \| P_U) < \delta_c$. A set C of at least two units is a candidate if it is a connected component of the coherence graph and if $D_{\mathrm{KL}}(P_U \| P_V) > \delta_w$ and $D_{\mathrm{KL}}(P_V \| P_U) > \delta_w$ for every $U \in C$ and every unit $V \notin C$. A handoff selects the union of the agents of C as a block once C has been a candidate without interruption for a time τ_p. A selected block is never dissolved.

The selector reads only the current presented laws and their recent history, so each handoff is a measurable function of accessible history, as (F2) requires. It is a declared threshold selector of the kind used in the Wishart, open-law and per-agent chapters of the main manuscript. Its block relation, internal coherence together with a wall, is the contrast structure that Item 3 found at stationary states with parents.

Definition 1.2 (Controls). The controls are those of the formation standard (definition 6.1). A cost is a peer cost when its compared law depends on the law of another agent; here these are the costs D_{ij} with $j \neq i$ and, after a handoff, the costs to the block's parent and to the other units. The label-preserving null Φ^0_{lp} deletes every peer cost while labels, priors and all other data are kept, so

the null self row is

$$\beta^0_{ii} = \frac{\pi_{ii} e^{-D_{ii}/T}}{\pi_{ii} e^{-D_{ii}/T} + 1 - \pi_{ii}}, \tag{1.2}$$

the convention of the per-agent chapter of the main manuscript (*Per-Agent Grouping with Persistent Records*). The renormalized null Φ^0_{rn} also removes the labels of the deleted costs, so the evaluator receives weight one and, in the Gaussian instance below, each agent follows $\dot{y}_i = (m_G - y_i) + \rho(o_i - y_i)$. The frozen-environment control Φ^{fr} deletes nothing and holds the law of every other agent at its initial value in every cost and row that enters the force on an agent. Along each control the selector reads the presented laws through the formulas of definition 1.1, and after a handoff along a control the control remains the fine flow, the selector reading the pools of the selected blocks computed from the fine states. The control without feedback Φ^{off} uses the same selector and records the same handoffs, but every agent keeps its individual row, sources and channels after a handoff, so the parents are passive.

The control without feedback is not part of the standard. It answers a different question: whether the action of a parent on its members changes what forms.

Remark 1.3 (Regularity and event conventions). In the Gaussian instance every flow used here, before, between and after handoffs and along every control, is the flow of a smooth vector field on $\mathbb{R}^{12}$ whose rows are smooth and bounded and whose growth is at most linear, so its solutions exist for all time and are unique; the same holds for the parent flow (iii) of theorem 4.1 on the unit means. For general laws no chart for the per-agent flow is declared; the general-law results below use only first variations and make no claim about flows. A moment projection onto a regular exponential family requires the pooled mean parameter to lie in the interior of the mean domain, which fails, for example, for a block of Bernoulli point masses. Candidates of definition 1.1 are connected components, hence disjoint, and all candidates whose windows end at one time are selected together. Where the derived selector of chapter 7 has several minimizers it takes the one whose blocks, listed by their smallest agents, come first lexicographically. The checks evaluate the selectors after every integration step of length 0.05 unless stated otherwise, so the event times they report lie on that grid.

Chapter 2

What top-down feedback cannot do

Proposition 2.1 (A parent as one more source). Let receiver i have the Gibbs row β_i over a finite source set $\mathcal{N}_i$ at temperature T, and add a source A with prior share $a \in (0,1)$, multiplying the other priors by $1-a$. The new row satisfies

$$\beta'_{i\nu} = (1-\beta'_{iA})\beta_{i\nu}, \qquad \nu \in \mathcal{N}_i. \tag{2.1}$$

If the first variation of D_{iA} vanishes at the current law of i, as it does when the compared law of A equals q_i, then the force on q_i is $1-\beta'_{iA}$ times its row part before the addition, plus the unchanged record force.

Proof. With $Z'_i = (1-a)Z_i + ae^{-D_{iA}/T}$ we have $\beta'_{i\nu} = (1-a)\pi_{i\nu}e^{-D_{i\nu}/T}/Z'_i = \beta_{i\nu}(1-a)Z_i/Z'_i$ and $1-\beta'_{iA} = (1-a)Z_i/Z'_i$, which is eq. (2.1). By the envelope identity the row part of $d\mathcal{F}'_i$ is $\sum_\nu \beta'_{i\nu}dD_{i\nu} + \beta'_{iA}dD_{iA}$, which equals $(1-\beta'_{iA})\sum_\nu \beta_{i\nu}dD_{i\nu}$ where $dD_{iA}=0$. The record term is weighted by no row, and the Fisher field is linear in the differential. □

The exponents of the row are unchanged, so the parent does not change which of the old sources the member trusts. Without records, the parent only rescales each member's velocity by a positive factor at agreement. With records, it raises the weight of the records relative to the row part by the factor $(1-\beta'_{iA})^{-1}$, a form of stubbornness. Away from agreement it also pulls the member toward its compared law, which reduces the spread inside the block. Check (2) confirms eq. (2.1) and the force decomposition. Check (15) runs twelve agents in four groups without records, each agent's evaluator target in its own row and each member adding its pooled parent with share a; for $a = 0$, 0.3 and 0.6 the same four blocks and the same two unions are selected, and the unions occur at $t \approx 40$, 100 and 245. Both nulls select nothing there, because the initial state is an equilibrium of each null flow, at which adjacent divergences are $0.27 > \delta_c$.

Proposition 2.2 (Top-down costs are internal forces). Let $\mathcal{E} = \mathcal{E}_0 + \tau\mathcal{E}_{\mathrm{td}}$ be continuously differentiable on a Riemannian manifold (S, g), and let $x(t)$ solve $\dot{x} = -\operatorname{grad}_g \mathcal{E}$. Let J be continuously differentiable on S with $\xi = \operatorname{grad}_g J$. If $\mathcal{E}_{\mathrm{td}}$ is invariant along ξ, that is $d\mathcal{E}_{\mathrm{td}}(\xi) = 0$, then

$$\frac{d}{dt}J(x(t)) = -d\mathcal{E}_0(\xi)\big(x(t)\big), \tag{2.2}$$

which does not involve τ.

Proof. $\frac{d}{dt}J(x) = dJ(\dot{x}) = g(\xi,\dot{x}) = -g(\xi, \operatorname{grad}_g \mathcal{E}) = -d\mathcal{E}(\xi) = -d\mathcal{E}_0(\xi)$. □

In the Gaussian location family with laws $\mathcal{N}(y_i, P_i^{-1})$, the Fisher metric on the means is $\bigoplus_i P_i$. The common translation of the members of a block I and of a parent A by $u \in \mathbb{R}^K$ is $\xi_u = \operatorname{grad} J_u$

with $J_u = u^T(\sum_{i\in I} P_i y_i + P_A y_A)$. The top-down factor of Item 3 depends on $y_i - y_A$ and on the cross-scale dictionary, so its cost is invariant under ξ_u, and the block momentum $\sum_{i\in I} P_i y_i + P_A y_A$ has a rate independent of τ at every configuration (check (3a)). Suppose in addition that the parent is the pool, with top-down cost $\frac{\tau}{2}\sum_{i\in I}\|y_i - \bar{y}_I\|^2$, that the members have equal precisions, and that every member of a block attends to every member of every other block with one weight. Then the block sums form a closed linear system, the external equitable partition of O'Clery et al. [17], and their trajectories do not depend on τ (check (3b), residual 6×10^{-16}). With sparse bridges, or with a parent that has its own state and prior, the block sums do depend on τ (checks (3b) and (3c)). The top-down cost moves momentum between the parent and the members and among the members, and the bridges read where it sits. The sharpening of walls under top-down coupling found in Item 3 is of this kind, and it is not available to a parent that is exactly inherited through the block momentum.

For general laws the proposition needs a momentum function for the collective motion. In natural coordinates of an exponential family the common shift of natural parameters has momentum $\sum_i \mu_i$, the sum of mean parameters. A Kullback–Leibler cost between members of the family is a Bregman divergence of the log-partition ψ, and differentiating it along a common shift shows that it is invariant for every pair exactly when $\nabla\psi$ is affine, that is, when the sufficient statistic is Gaussian with one covariance. Such families are Gaussian location families in the coordinates of the sufficient statistic; the common-variance lognormal family, with sufficient statistic $\log y$, is one. Outside them the proposition applies to costs constructed to be invariant, not to the divergences of the theory.

Proposition 2.3 (A globally attracting equilibrium excludes counterfactual recursion under all-time persistence). Fix the first-stage blocks of a construction and let $\mathcal{C}$ be a set of post-handoff states. Suppose the post-handoff flow is deterministic and every trajectory starting in $\mathcal{C}$ converges to one equilibrium z^*. Let the relation that some second-stage block B is a candidate be given by strict inequalities between continuous functions of the state, defining an open set S, and let persistence mean that the relation holds at all later times. If $z^* \in S$ and no other block is selected along these trajectories before B, the block B is selected from every state in $\mathcal{C}$ and (F6b) fails for B. If $z^* \notin \bar{S}$, no state in $\mathcal{C}$ leads to a persistent selection of B.

Proof. If $z^* \in S$, some neighborhood of z^* lies in S, and every trajectory from $\mathcal{C}$ remains in it after a finite time, so the candidate persists beyond any window τ_p; since no other selection intervenes, the flow is unchanged until then. If the relation holds at all times after some t_0 along a trajectory converging to z^*, then z^* is a limit of points of S. □

The proposition says nothing when $z^* \in \partial S$: for $\dot{x} = -x$ on $\mathbb{R}$ and $S = \{x > 0\}$, trajectories from $x_0 > 0$ stay in S and those from $x_0 < 0$ never enter it. Without the hypothesis on competing selections the first alternative fails when another block is selected during a transient and changes the flow. The selector of definition 1.1 persists for a finite window, not at all later times, and for it the proposition gives no conclusion about (F6b).

Remark 2.4 (A convex hierarchy with a finite dwell meets (F6b) (numerical)). Take two first-stage blocks $I = \{a_1, a_2\}$ and $J = \{b_1, b_2\}$ with records $o = (-0.775, -0.725, 0.725, 0.775)$, and add at the handoff declared parents with means m_I, m_J, initialized at the pools of their members, with

the strictly convex energy

$$\mathcal{E} = \sum_{B \in \{I,J\}} \Big[\sum_{i \in B} \big(\tfrac{1}{2}(y_i - o_i)^2 + \tfrac{1}{2}(y_i - m_B)^2 \big) + 4 m_B^2 \Big], \tag{2.3}$$

whose last terms anchor both parents at zero. Let the flow be its gradient flow in the constant metric $\mathrm{diag}(20, 20, 20, 20, 6, 6)$, and let the selector of definition 1.1 read the parents' presented laws of precision 6, with $\delta_c = 0.05$, $\delta_w = 0.5$ and $\tau_p = 2$. The Hessian has eigenvalues 1.757, 1.757, 2, 2, 10.24 and 10.24, and the unique minimizer has parent gap $1/6$, with divergence $0.083 > \delta_c$, outside the closure of the coherence set. From a handoff with block gap 0.5, at which the members of each block are coherent and the blocks are walled, the parent gap falls to 0.1175, stays below the coherence radius for 4.81 time units, and the union is selected at $t = 3.785$. From a handoff with block gap 0.7, also walled, the parent gap stays above 0.150 and no union forms. Both trajectories converge to the minimizer (check (29)). The length of a transient dwell in the candidate set depends on the handoff, and that suffices for (F6b) relative to these two handoffs. In the Gaussian tangent global joint of Items 2 and 3, whose own flow uses the Fisher metric of its factors, this energy comes with the metric $\mathrm{diag}(2, 2, 2, 2, 10, 10)$, and then neither handoff unites. A random search over that joint, with first stages produced by its own flow and the selector of definition 1.1, found instances meeting (F6b), every one of them with declared parent anchors that pull the parents together, and none with each parent anchored at the record mean of its own cluster; this search is not among the checks. Whether declared parent anchors count as new constitutive data under (F6a), and whether such instances exist when parents interact only through inherited bridges, are open (open problem 8.4).

Chapter 3

Shared labels

Definition 3.1 (Shared-label handoff). Let a handoff select a block I of $n = |I| \geq 2$ agents. Suppose that every member compares each source outside I through the same channel, so that a source ν presents one compared law ν_ν to all members, and that the members share the evaluator G. From the handoff on:

(a) The members carry one source label J_I. Its sources are the evaluator, a parent A_I, and the units outside I. Its prior π_I is the average of the members' priors, with the mass that the members gave to one another assigned to A_I and the mass that they gave to the agents of an outside unit assigned to that unit. Each member descends, with respect to its own law and with all other laws as data, the block free energy

$$\mathcal{F}_I = \sum_\nu \beta_{I\nu} \sum_{i\in I} D_{i\nu} + T D_{\mathrm{KL}}(\beta_I \| \pi_I) + \sum_{i\in I} \int -\log \ell_i(o_i \mid k) q_i(dk), \tag{3.1}$$

whose minimizing row is $\beta_{I\nu} = \pi_{I\nu} \exp(-\sum_{i\in I} D_{i\nu}/T)/Z_I$.

(b) The parent presents the linear pool $\bar{q}_I = n^{-1} \sum_{i\in I} q_i$ or, when the compared laws lie in a declared family E, its moment projection P_I onto E. The members compare themselves to it, $D_{iA} = D_{\mathrm{KL}}(q_i \| P_I)$, and every receiver outside I replaces the agents of I in its source set by A_I, with their total prior.

(c) A later handoff that selects a set of units replaces them in the same way by one block whose members are all their agents. The parents of the replaced blocks are then no longer sources.

The tempered label replaces T by nT in the row of (a).

Part (a) is top-down attention: a member no longer decides alone which source to trust; the block decides, with every member's evidence. Part (b) is top-down prediction and routing: members are pulled toward what the block presents, and the block speaks to the rest of the population through its parent. The rule is declared once, before any selection, and applies at every level, so a second stage uses it without a new parameter.

Proposition 3.2 (The shared row reads the pool). Under definition 3.1, for every source ν whose costs are finite,

$$\sum_{i\in I} D_{\mathrm{KL}}(q_i \| \nu_\nu) = \sum_{i\in I} D_{\mathrm{KL}}(q_i \| \bar{q}_I) + n D_{\mathrm{KL}}(\bar{q}_I \| \nu_\nu). \tag{3.2}$$

Hence

$$\beta_{I\nu} = \frac{\pi_{I\nu} \exp\big(-n D_{\mathrm{KL}}(\bar{q}_I \| \nu_\nu)/T\big)}{\sum_\mu \pi_{I\mu} \exp\big(-n D_{\mathrm{KL}}(\bar{q}_I \| \nu_\mu)/T\big)}, \tag{3.3}$$

the Gibbs row at temperature T/n of a single receiver with law $\bar{q}_I$, and the parent presenting $\bar{q}_I$ has relative cost zero. If every compared law lies in a regular exponential family E with sufficient statistic t, and the parent presents the moment projection P_I, defined by $\mathbb{E}_{P_I} t = \bar{\mu}_I = n^{-1} \sum_i \mathbb{E}_{q_i} t$, then for every $p \in \mathsf{E}$

$$D_{\mathrm{KL}}(\bar{q}_I \| p) = D_{\mathrm{KL}}(\bar{q}_I \| P_I) + D_{\mathrm{KL}}(P_I \| p), \tag{3.4}$$

and the shared row is the Gibbs row at temperature T/n of the costs $D_{\mathrm{KL}}(P_I \| \nu_\nu)$. It depends on the block only through $\bar{\mu}_I$, and A_I again has relative cost zero.

Proof. Write $\log(q_i/\nu) = \log(q_i/\bar{q}_I) + \log(\bar{q}_I/\nu)$, integrate against q_i, and sum over i, using $\sum_i q_i = n\bar{q}_I$; this is the receiver-pool identity of the dynamics chapter with equal weights. The term $\sum_i D_{\mathrm{KL}}(q_i \| \bar{q}_I)$ is the same for every source and cancels from the normalized row, and with the parent presenting $\bar{q}_I$ it is the whole cost of A_I. For eq. (3.4), write $p = p_\theta$ with log density $\theta \cdot t - \psi(\theta)$ relative to a base measure; then $D_{\mathrm{KL}}(\bar{q}_I \| p_\theta) = -H(\bar{q}_I) - \theta \cdot \bar{\mu}_I + \psi(\theta)$ with H the entropy relative to that measure. Subtracting the same expression at θ^* with $\nabla\psi(\theta^*) = \bar{\mu}_I$ gives $\psi(\theta) - \psi(\theta^*) - (\theta - \theta^*) \cdot \nabla\psi(\theta^*) = D_{\mathrm{KL}}(p_{\theta^*} \| p_\theta)$, the Pythagorean identity of Csiszár [2]. The cost of A_I is then the common term plus $nD_{\mathrm{KL}}(\bar{q}_I \| P_I)$, which is also common to all sources. □

Check (1) confirms eqs. (3.2) to (3.4) on a six-point alphabet, including that two blocks with the same pool and different members have the same row.

Remark 3.3 (When the shared row is a posterior). In the registered-agreement model of the dynamics chapter, an agent's Gibbs row is the posterior of the source it heard, given that it registered agreement with that source, at temperature equal to the reciprocal message count. If the members of I heard one common source J_I and their registrations are conditionally independent given J_I, the posterior of J_I is π_I times the product of the members' registration probabilities, and the source-independent factors cancel; it is the row of definition 3.1(a). The shared row is then the exact posterior of a common cause. Members that have influenced one another do not in general supply conditionally independent evidence, and the row is then a Gibbs row of summed losses, a generalized-Bayes update in the sense of Bissiri et al. [1], not a posterior. It adds the members' costs, unlike a logarithmic opinion pool with weights summing to one [6]; the tempered label is the pool that averages them.

Proposition 3.4 (A tempered label is neutral at synchrony). Suppose the priors are homogeneous, $\pi_{ii} = \pi_s$ and $\pi_{ij} = \pi_p$ for $j \neq i$, all channels are identity channels, and every parent presents the linear pool of its members, or every member law lies in the family E onto which the parents project. At a configuration where every block is internally synchronized, $q_i = \bar{q}_I$ for $i \in I$, the force on every agent under the shared-label handoff with the tempered label equals its force under the control without feedback. With the untempered label, every cost in the row of a block of n members is n times the corresponding cost in the control.

Proof. At synchrony, in the control, member i of I has the evaluator cost $D_{\mathrm{KL}}(\bar{q}_I \| G)$, cost zero to each of the other $n-1$ members with total prior $(n-1)\pi_p$, and cost $D_{\mathrm{KL}}(\bar{q}_I \| \bar{q}_V)$ to each of the n_V agents of an outside unit V, with total prior $n_V\pi_p$; the sources at cost zero exert no force. The tempered shared row has costs $n^{-1} \sum_{i \in I} D_{i\nu}$, which at synchrony are $D_{\mathrm{KL}}(\bar{q}_I \| G)$, zero for A_I with prior $(n-1)\pi_p$, and $D_{\mathrm{KL}}(\bar{q}_I \| P_V) = D_{\mathrm{KL}}(\bar{q}_I \| \bar{q}_V)$ for V with prior $n_V\pi_p$, because under either

hypothesis $P_V = \bar{q}_V$ at synchrony. The rows, the compared laws and the record forces coincide. Without tempering each cost is the sum of n equal member costs. □

Check (6) gives a difference of 2×10^{-16} between the tempered label and the control at a synchronized configuration, and 0.06 for the untempered label. At synchrony the whole effect of the shared label is evidence accumulation: a block of n members is n times as selective as one of its members. The synchronized set is not invariant when the records differ inside blocks, so the proposition compares vector fields there, not trajectories; the tempered label differs from the control only through the spread inside blocks.

Remark 3.5 (Dictionaries). With receiver-owned dictionaries, as in Items 1 to 3, member i reads source ν through its own dictionary $\bar{\theta}_{i\nu}$, acting by push-forward, and the compared laws differ across members. Relative entropy is invariant under one measurable bijection applied to both arguments, so if there are measurable bijections g_i, applied to the members' laws, with $g_i \bar{\theta}_{i\nu} = \Theta_\nu$ for every source ν outside I, then $\sum_i D_{\mathrm{KL}}(q_i \| \bar{\theta}_{i\nu}\nu) = \sum_i D_{\mathrm{KL}}(g_i q_i \| \Theta_\nu \nu)$, and proposition 3.2 applies to the laws $g_i q_i$. Such maps exist exactly when $\bar{\theta}_{j\nu}\bar{\theta}_{i\nu}^{-1}$ does not depend on ν for all members i, j: then $g_j = \bar{\theta}_{1\nu}\bar{\theta}_{j\nu}^{-1}$ works for every ν, and conversely $g_j^{-1} g_i = \bar{\theta}_{j\nu}\bar{\theta}_{i\nu}^{-1}$. The maps g_i act on the members' laws at fixed sources; they are not passive frame changes. Equivalently, every closed walk that goes from member i to an outside source ν, from ν to member j, from j to a second source μ and from μ back to i, with each source edge traversed against its direction by the inverse of its dictionary, has trivial signed-walk holonomy, and in this sense the block is flat toward its sources, with Θ_ν as a common meta-level dictionary. This is not the condition of the flat-wall proposition of Item 3, which concerns directed walks and needs internal directed paths and bridges in both directions. For a block that is curved toward its sources the shared row is not in general a function of the pool of any such images of the members' laws; it can be when the member laws are invariant under the discrepancies, as isotropic Gaussian laws are under rotation dictionaries.

Chapter 4

Exact parents in the Gaussian location family

Theorem 4.1 (Exact inheritance of shared-label parents at a fixed partition). Let $\mathsf{Y} = \mathbb{R}^K$, let every agent law, compared law, presented law and the evaluator be Gaussian with one fixed precision matrix Λ, recognition being restricted to the location family $\mathcal{N}(y, \Lambda^{-1})$ whose Fisher metric on the means is Λ, let all channels be identity channels, and let each record have the likelihood $\mathcal{N}(o_i; k, R^{-1})$; write $\rho = \Lambda^{-1}R$. The parent of a block presents $\mathcal{N}(\bar{y}_U, \Lambda^{-1})$, the moment projection of its pool onto the location family. After any sequence of shared-label handoffs, let $\mathcal{U}$ be the set of units, n_U their sizes, $\bar{y}_U$ and $\bar{o}_U$ the averages of the members' means and records, and m_ν the mean of source ν: m_G for the evaluator, $\bar{y}_U$ for the unit's own parent, and $\bar{y}_V$ for a unit $V \neq U$. Then for every $U \in \mathcal{U}$ and $i \in U$:

(i) $\beta_{U\nu} = \pi_{U\nu} \exp\!\big(-\frac{n_U}{2T}\|\bar{y}_U - m_\nu\|^2_\Lambda\big)/Z_U$;

(ii) $\dot{y}_i = \sum_\nu \beta_{U\nu}(m_\nu - y_i) + \rho(o_i - y_i)$;

(iii) $\frac{d}{dt}\bar{y}_U = \sum_\nu \beta_{U\nu}(m_\nu - \bar{y}_U) + \rho(\bar{o}_U - \bar{y}_U)$;

(iv) $\frac{d}{dt}(y_i - \bar{y}_U) = -(I + \rho)(y_i - \bar{y}_U) + \rho(o_i - \bar{o}_U)$.

Consequently, on every set of configurations with one fixed set of units that is invariant under the flow (ii), that is, between handoffs, the retention $R(x) = (\bar{y}_U)_{U\in\mathcal{U}}$ satisfies the law condition (L), the dynamical condition (D) with parent flow (iii), the channel condition (C), and the response condition (B) for the read-only parent protocols of definition 6.2 answered within the same interval, as these conditions are defined for an inherited parent in the inheritance chapter. The retention is not injective when some unit has $n_U \geq 2$, and two configurations in one fiber remain distinct and in one fiber at all later times of the interval.

Proof. (i) For a source with mean m,

$$\sum_{i\in U} D_{\mathrm{KL}}\big(\mathcal{N}(y_i, \Lambda^{-1}) \| \mathcal{N}(m, \Lambda^{-1})\big) = \frac{n_U}{2}\|\bar{y}_U - m\|^2_\Lambda + \frac{1}{2}\sum_{i\in U}\|y_i - \bar{y}_U\|^2_\Lambda, \tag{4.1}$$

and the second term, which is the whole cost of the own parent, does not depend on the source; it cancels from the normalized row. (ii) The derivative of eq. (3.1) with respect to y_i is $\sum_\nu \beta_{U\nu}\Lambda(y_i - m_\nu) + R(y_i - o_i)$, since the record term is $\frac{1}{2}\|o_i - y_i\|^2_R$ plus a constant. The Fisher flow multiplies by $-\Lambda^{-1}$, and the rows sum to one. (iii) Average (ii) over $i \in U$; the rows and the source means do not depend on i. (iv) Subtract (iii) from (ii) and use $\sum_\nu \beta_{U\nu} = 1$.

By (i) the rows of every unit are functions of the unit means. By the routing of definition 3.1(b) every compared law is the evaluator or a presented law $\mathcal{N}(\bar{y}_V, \Lambda^{-1})$, so (iii) is a closed system in $(\bar{y}_U)_U$

and $R \circ \Phi_t = \Phi^c_t \circ R$ for the flow Φ^c of (iii), which is well posed by remark 1.3. The presented laws, and the rows, channels and evaluator read by (iii), are functions of R, which gives (L) and (C). The deviations $(y_i - \bar{y}_U)_{i \in U}$ with zero sum span a subspace of dimension $(n_U - 1)K$ on which R is constant. By (iv) they evolve by an invertible affine flow that does not depend on the unit means, so distinct deviations stay distinct while the unit means move together by (iii).

For (B), a read-only query of A_U at time s returns a draw from $\mathcal{N}(\bar{y}_U(s), \Lambda^{-1})$, a function of $R(\Phi_s x) = \Phi^c_s(R(x))$, and it changes no state, record or row. The joint law of the response and of the retained state at any later time s' of the interval is therefore $K\big(\Phi^c_s R(x), \cdot\big) \otimes \delta_{\Phi^c_{s'} R(x)}$, with $K(z, \cdot)$ the presented law at retained value z, a function of $R(x)$. For a finite adaptive schedule, in which the time and the target of each query are measurable functions of the earlier responses, the joint law of the responses and the retained state factors through $R(x)$ by induction on the number of queries, because the responses do not enter the dynamics. □

Check (7) fixes the four first-stage units of the twelve-agent instance of chapter 5 after their handoffs and integrates the fixed-partition flow (ii) for twenty time units; it does not run the selector, and the instance itself unites inside that window. The block means agree with the solution of (iii) to 8×10^{-15}, and the deviations agree with the closed form of (iv) to 1.4×10^{-15}. Exact block-mean dynamics of linear consensus over external equitable partitions on a fixed graph were analyzed by Martini et al. [13] and Egerstedt et al. [4] and related to observability and coarse graining by O'Clery et al. [17], and for nonlinear networks the quotient over such partitions describes the synchronized states [20]. Here the rows depend on the state, the closure holds for every spread inside the blocks, and it is created by the handoff rule.

Remark 4.2 (What is not proved: the hybrid process). Theorem 4.1 is a statement at a fixed partition. The process to which (F3) applies is hybrid. The selector runs candidate clocks, and the first-stage handoffs are not simultaneous: on the evaluation grid of the checks they occur at four times between $t = 2.13$ and $t = 2.35$. At the first handoff the clocks of the other groups are running. They started under the dynamics with individual rows, which depend on the members' individual laws, so they are not functions of the retained means of a block already handed off, and they fix the later handoff times. The mean-only retention also omits the partition, the block sizes n_U, which enter the union's temperature and its prior masses, the record means $\bar{o}_U$ and the reassigned prior rows π_U, and neither the merge of retained states at the union nor the descent of the selection events is proved. Condition (B) is proved only for read-only queries. A protocol that assimilates a member-specific record fails it: a record o' with Gaussian likelihood of precision R' sent to member k moves the block mean by $n^{-1}(\Lambda + R')^{-1}R'(o' - y_k)$, which is not a function of R. Clause (F3) for the hybrid process is therefore open; a route is stated in remark 4.4.

Corollary 4.3 (Composition and type at a fixed partition). The parent flow (iii) is the per-agent flow of a single Gaussian location agent with mean $\bar{y}_U$ and record $\bar{o}_U$, at temperature T/n_U, whose source set contains, besides the evaluator and the other units, a source at its own law with prior π_{UA}. A shared-label handoff applied by the primitive rule of definition 3.1(c) to units formed by earlier handoffs returns a unit of the same form whose size is the sum of their sizes, and at a fixed partition successive retentions compose by the composition proposition of the inheritance chapter. A block formed from blocks I and J has temperature $T/(n_I + n_J)$, and the map from the two parents to their union, $(\bar{y}_I, \bar{y}_J) \mapsto (n_I \bar{y}_I + n_J \bar{y}_J)/(n_I + n_J)$, is again not injective.

Proof. The first statement is (iii) read as the flow (ii) of a single agent with the stated rows; a single agent is the case $n_U = 1$, with prior zero on the source at its own law. A handoff by the primitive rule adds the members' costs, which by (i) are the units' costs weighted by their sizes, and returns a unit of the stated form with the summed size. At a fixed partition, composition is the composition of the two exact retentions. □

The corollary does not define a handoff on agents of this class: the rule of definition 3.1 is stated for primitive agents with one temperature. Treating each parent as one agent whose cost is counted once, with priors averaged without weights, does not in general reproduce the primitive rows for blocks of unequal size; the rule that does adds the exponents $D_{\mathrm{KL}}(P_U\|\nu_\nu)/T_U$ with $T_U = T/n_U$ and averages priors and record means with weights n_U (remark 4.4). Composition across a union also needs the merge of retained states to descend, which is not proved.

Remark 4.4 (A mass-carrying parent). The following is a route to (F3) for the hybrid process, not a result of this document. Let the parent of a unit carry $p_U = (n_U, \bar{y}_U, \bar{o}_U, \pi_U)$, with Gibbs exponents $n_U D_{\mathrm{KL}}(P_U\|\nu_\nu)/T$, temperature T/n_U and Fisher metric Λ. Merge units $U_1, \dots, U_m$ by $n = \sum_k n_k$, $\bar{y} = \sum_k n_k \bar{y}_k/n$, $\bar{o} = \sum_k n_k \bar{o}_k/n$ and $\pi = \sum_k n_k \pi_k^{\mathrm{re}}/n$, where π_k^{re} moves the mass that U_k gave to the other merged units onto the new parent. Associativity is that of weighted means, and agreement with the primitive rule would follow from the blockwise identity

$$\sum_{i\in W} D_{\mathrm{KL}}(q_i\|\nu) = \sum_k \Big[\sum_{i\in U_k} D_{\mathrm{KL}}(q_i\|P_{U_k}) + n_k D_{\mathrm{KL}}(P_{U_k}\|\nu)\Big], \qquad W = \bigcup_k U_k, \tag{4.2}$$

which holds for moment-projected location laws by the decomposition of a sum of squares. Add the running candidate clocks to the retained state. Both selectors read only presented laws, so the selection events, the clocks and the merges would then be functions of the retained state, and with the read-only class of definition 6.2 the joint kernels would factor as in the proof of theorem 4.1. Whether the retained state also records formation times decides the second-stage verdict of (F4) (chapter 6 and open problem 8.2).

Theorem 4.5 (Exact pooled parents force a Gaussian sufficient statistic). Let the members' laws and all compared laws lie in a regular exponential family with sufficient statistic $t : \mathsf{Y} \to \mathbb{R}^d$, natural parameter θ in an open set Θ, log-partition ψ, and mean parameter $\mu = \nabla\psi(\theta)$. Let the dynamics be the per-agent Fisher flow of the block free energy eq. (3.1) without records, with the parent presenting the moment projection of the pool. Suppose that for blocks of two members the velocity of the pooled mean parameter $\bar{\mu} = (\mu_1 + \mu_2)/2$ is, at every configuration, a function of $\bar{\mu}$ and of the row-weighted source parameter $c = \sum_\nu \beta_\nu \theta_\nu$, for every value of c in Θ. Then $\nabla^2\psi$ is constant on Θ, and the laws of t form a Gaussian family with one covariance and varying mean; when t is the identity on $\mathsf{Y} = \mathbb{R}^d$, the family is the Gaussian location family.

Proof. In natural coordinates the Fisher flow is $\dot{\theta}_i = -\partial_{\mu_i}\mathcal{F}_I$, and $\partial_\mu D_{\mathrm{KL}}(p_\theta\|p_{\theta'}) = \theta - \theta'$, so $\dot{\theta}_i = c - \theta_i$, the rows summing to one. With $H(\mu) = \nabla^2\psi(\theta(\mu))$, the covariance of t, the pooled velocity is $\frac{1}{2}\sum_i H(\mu_i)(c - \theta(\mu_i))$. It is affine in c. Requiring it to be constant on each fiber $\{(\mu_1, 2\bar{\mu} - \mu_1)\}$ for every c in the open set Θ makes both $H(\mu_1) + H(2\bar{\mu} - \mu_1)$ and $K(\mu_1) + K(2\bar{\mu} - \mu_1)$ independent of μ_1, where $K(\mu) = H(\mu)\theta(\mu)$. The mean-parameter domain is open and convex. The first condition gives $DH(\mu_1) = DH(\mu_2)$ for all μ_1, μ_2, so H is affine. Since $D\theta = H^{-1}$,

$DK(\mu)[v] = DH[v]\theta(\mu) + v$, and the second condition makes $DH[v]\theta(\mu)$ independent of μ; as $\theta(\mu)$ ranges over the open set Θ, $DH[v] = 0$ for every v. Hence $\nabla^2\psi$ is constant and ψ is quadratic. The Laplace transform of the law of t under the base measure is e^{ψ}, so by its uniqueness the laws of t are Gaussian with covariance $\nabla^2\psi$; for $d = 1$ this is the constant-variance case of the classification of Morris [15]. □

For Bernoulli laws $H(p) = p(1-p)$. Two members with $p = (0.5, 0.5)$ and $p = (0.2, 0.8)$ have the same pool and the same rows, and with a single source of natural parameter 1 their pooled velocities are 0.25 and 0.16 (check (4)); no update of the pool reproduces both. The theorem concerns the pooled parent only. Another retention, such as the canonical parent of the inheritance chapter for the routed interface, may still be exact for general laws; whether it compresses is open (open problem 8.1).

Remark 4.6 (What the parent forgets). The fibers of R are transient. By (iv) each member's deviation converges at the rates of $I + \rho$ to $(I + \rho)^{-1}\rho(o_i - \bar{o}_U)$, so on the attracting set each member's law is determined by its parent's mean and its own record. The parent forgets the members' records except their mean, and the transient part of their deviations; it forgets nothing that the members' own records do not restore. The fibers are inert in the sense used in the Wishart chapter of the main manuscript (*Two-Stage Selection with Exact Inheritance on a Wishart Family*): no presented law, row or record depends on them. They are not orbits of a symmetry of the fine field, since a shift of the deviations does not preserve the affine equation (iv). Relative to the child-level probes of definition 6.2 they are not gauge, because a receiver attending to one member distinguishes two configurations of a fiber; relative to the parent protocols they are invisible. The interface on which R is exact does not exist before the handoff. While the members attend individually, their Gibbs rows, and the rows of every receiver attending to them, depend on each member's own mean, and the block mean is not autonomous. The handoff creates this operational equivalence; it is not present for every partition beforehand, which distinguishes the construction from the quotients supplied before membership in the open-law and per-agent chapters of the main manuscript. It is created by a declared rule, however, not by dynamics that are unchanged at the handoff.

Chapter 5

An instance

Take $N = 12$, $K = 1$, $\Lambda = \lambda = 6$, $T = 1$, homogeneous priors $\pi_s = 0.05$ and $\pi_p = 0.95/11$, evaluator mean $m_G = 0$, and $\rho = 0.005$. The records form four groups of three, $L_1 = (-3.2, -3.05, -2.9)$, $L_2 = (-2.1, -1.95, -1.8)$, $R_1 = -L_2$ and $R_2 = -L_1$, held by agents 0 to 2, 3 to 5, 6 to 8, and 9 to 11. The thresholds are $\delta_c = 0.05$, $\delta_w = 0.5$ and $\tau_p = 2$. Every cost is $\frac{\lambda}{2}$ times a squared difference of means, so two units are coherent when their means differ by less than 0.129 and walled when they differ by more than 0.408. The admitted class is

$$\mathcal{A} = \{y(0) \in \mathbb{R}^{12} : |y_i(0) - o_i| < 0.005 \text{ for all } i\}, \tag{5.1}$$

which is open. No pair is coherent at $t = 0$, since adjacent records differ by 0.15 and $\frac{\lambda}{2}(0.14)^2 = 0.0588 > \delta_c$. The members of a group disagree in their records, so a group becomes coherent only if the interaction contracts it; this is the device of the per-agent chapter of the main manuscript. The flows are those of theorem 4.1(ii) after the handoffs and of the per-agent Gibbs flow with individual rows before them.

Proposition 5.1 (The null flow never selects). From every state in $\mathcal{A}$ the null flow never selects a block, so its partition sequence is the initial partition alone.

Proof. Under Φ^0 each agent follows $\dot{y}_i = \beta^0_{ii}(y_i)(m_G - y_i) + \rho(o_i - y_i)$ with β^0_{ii} of eq. (1.2) and $D_{ii} = \frac{\lambda}{2}y_i^2$, so $\beta^0_{ii}(y) \le \frac{\pi_s}{1-\pi_s}e^{-3y^2}$. On $|y| \ge 1.795$ this is at most 3.3×10^{-6}, and the evaluator term is at most 1.1×10^{-5} in absolute value for $|y| \le 3.205$. At $y = o_i \pm 0.005$ the record term has magnitude 2.5×10^{-5} and points toward o_i, so the interval $[o_i - 0.005, o_i + 0.005]$ is forward invariant. Every pair of agents then stays at distance at least 0.14, with divergence at least $0.0588 > \delta_c$, so no pair is ever coherent and no candidate exists. □

Check (8) integrates the null flow to $t = 1000$: the agents end within 0.0012 of their records, and the smallest divergence between adjacent agents is 0.0675.

Proposition 5.2 (The renormalized null selects only the whole population). From every state in $\mathcal{A}$ the renormalized null flow selects no block before the block $\mathcal{V}$ of all twelve agents, which it selects once; its partition sequence is the initial partition followed by $\{\mathcal{V}\}$.

Proof. Under Φ^0_{rn} the agents follow $\dot{y} = -\kappa y + \rho o$ with $\kappa = 1 + \rho$, since $m_G = 0$. For agents $i \ne j$ write $o_{ij} = o_i - o_j$, $y_{ij} = y_i - y_j$, $r = \rho/\kappa$ and $e(t) = e^{-\kappa t}$. Then

$$y_{ij}(t) = o_{ij}c(t) + \varepsilon_{ij}e(t), \qquad c(t) = r + (1 - r)e(t), \qquad \varepsilon_{ij} = y_{ij}(0) - o_{ij}, \tag{5.2}$$

with $|\varepsilon_{ij}| < 0.01$ on $\mathcal{A}$. Since $c \ge (1 - r)e$ and $|o_{ij}| \ge 0.15$, $y_{ij} = o_{ij}c(1 + \theta_{ij})$ with $|\theta_{ij}| \le 0.01/(0.15(1 - r)) < 0.0671$. The agents therefore keep the order of their records, a coherent pair,

at distance below 0.12910, has $c|o_{ij}| < 0.1384$, and a walled pair, at distance above 0.40824, has $c|o_{ij}| > 0.3826$. Before any handoff the units are the agents, and a candidate other than $\mathcal{V}$ is a set of consecutive agents whose adjacent record spacings s satisfy $cs < 0.1384$ and whose adjacent outside spacings g satisfy $cg > 0.3826$, so that $g > 2.76s$ for the largest inner spacing s. The record spacings are 0.15 within groups, 0.8 between the two groups of a side and 3.6 between the sides, so the only such sets are the four groups, which require $c \in (0.3826/0.8, 0.1384/0.15)$, and the two sides, which require $c \in (0.3826/3.6, 0.1384/0.8)$. Because $c - r = (1 - r)e^{-\kappa t}$ decreases, the time spent with c in an interval (c_1, c_2) is $\kappa^{-1} \log((c_2 - r)/(c_1 - r))$, at most 0.659 for the groups and 0.504 for the sides, both below $\tau_p = 2$, so neither is selected. Every adjacent pair is coherent once $3.6c < 0.1384$, which happens in finite time because $3.6r < 0.018$, and it stays coherent because c decreases. With no unit outside it, $\mathcal{V}$ is then a candidate at all later times and is selected after τ_p; after that handoff no candidate remains. □

Check (28) confirms the history $\{\mathcal{V}\}$ at $t = 5.55$ on the evaluation grid from the fixed draw and twelve further states of $\mathcal{A}$.

Remark 5.3 (Selections of the shared-label flow (numerical)). From every tested state in $\mathcal{A}$, fifteen in all (the fixed draw of checks (7) to (10), twelve uniform draws in check (12), and one draw each in checks (11) and (14)), the shared-label flow selects L_1, L_2, R_1 and R_2, and then $L_1 \cup L_2$ and $R_1 \cup R_2$; on the evaluation grid the first-stage handoffs fall between $t = 2.13$ and $t = 2.35$ and the unions between $t = 6.95$ and $t = 8.05$. The selected blocks are the same for every state. Along the fixed-draw trajectory each group enters the candidate condition transversally, the slope of its coherence margin being at least 0.092 at entry, and over the rest of its persistence window the coherence margin, δ_c minus the largest edge of the group's minimum spanning tree, stays at least 0.0045 while the wall margin stays at least 0.43 (check (17)). The recorded gaps between the left parents after the first stage, from two states, lie between 0.595 and 0.622, at least 0.33 below $d_u - 0.005$ (checks (9), (10) and (14)), so by corollary 5.5 the unions follow from certified inequalities, and only the first stage rests on integration. By continuous dependence on initial data, which holds across the switching times of the handoffs because each entry is transversal, the same blocks are selected on an open neighborhood of the fixed draw, provided the integration error on $[0, 2.6]$ stays below these margins. The integration is fourth-order Runge–Kutta with fixed steps between 0.005 and 0.05; it was not validated by interval arithmetic.

Proposition 5.4 (The second stage reduces to a gap equation). After the four first-stage handoffs write $m_1 = \bar{y}_{L_1}$, $m_2 = \bar{y}_{L_2}$, $d = m_2 - m_1$, and $s = \bar{o}_{L_2} - \bar{o}_{L_1}$, which is 1.1 in the instance; the same statement holds for the variants of remark 5.6 with record gap $s \in [1.0, 1.3]$, whose left record means lie in $[-3.15, -1.85]$. While the left parent means lie in $[-3.3, -1.7]$ and the right ones in $[1.7, 3.3]$,

$$\dot{d} = F(d) + r(t), \qquad F(d) = \rho(s - d) - 2w_3(d)d, \qquad w_n(d) = \frac{ne^{-ad^2}}{n - 1 + ne^{-ad^2}}, \tag{5.3}$$

with $a = n\lambda/(2T) = 9$ for $n = 3$ and $|r(t)| \le 10^{-10}$.

Proof. By theorem 4.1(i) the row of L_1 has the evaluator with prior π_s and cost $9m_1^2$, its own parent with prior $2\pi_p$ and cost zero, L_2 with prior $3\pi_p$ and cost $9d^2$, and R_1, R_2 with prior $3\pi_p$

each and cost $9(m_1 - \bar{y}_R)^2$. On the stated ranges the evaluator weight relative to $2\pi_p$ is at most $\frac{\pi_s}{2\pi_p} e^{-26.0} < 1.5 \times 10^{-12}$, and the right-side weights are below e^{-100}. Dropping these weights from (iii) leaves $\dot{m}_1 = w_3(d)(m_2 - m_1) + \rho(\bar{o}_{L_1} - m_1)$, the prior π_p cancelling from w_3, and the same for m_2 with the roles exchanged; the dropped terms contribute less than 10^{-10} to $\dot{d}$. Subtracting the two equations gives eq. (5.3). The stated ranges are forward invariant up to these terms, since each parent is pulled toward the other left parent and toward its own record mean, both inside the range. □

For $s = 1.1$ the function F has exactly three zeros in $(0, \infty)$, $d_m = 0.0045647$, $d_u = 0.9630362$ and $d_s = 1.0831195$, each enclosed in an interval of width below 10^{-9}, with $F'(d_m) = -1.205$, $F'(d_u) = 0.0062$ and $F'(d_s) = -0.0034$ (check (9)). Check (27) certifies the zero count and the enclosures by interval arithmetic, using that both terms of F are negative for $d \geq s$, together with the bounds $F' > 0.005$ within 0.005 of d_u, $F' < -0.002$ within 0.005 of d_s, $F < -3.2 \times 10^{-5}$ on $[0.12909, d_u - 0.005]$, $F > 1.6 \times 10^{-5}$ on $[d_u + 0.005, d_s - 0.005]$ and $F < -1.7 \times 10^{-5}$ on $[d_s + 0.005, 1.6]$. The merged state d_m and the separated state d_s are stable, and the middle root d_u separates their basins. The separated state is an equilibrium held by the records: without them, $\rho = 0$, the only zero is $d = 0$. For $s = 1.0722$ the equation has one zero and for $s = 1.0723$ three (check (27)), so the middle root and the separated state appear in a saddle-node between these record gaps.

Corollary 5.5 (Shifted post-handoff states). Let $s = 1.1$ and consider post-handoff states in which L_1, L_2, R_1, R_2 are the selected blocks, the left parent means lie in $[-3.3, -1.7]$ with gap d_0, and the right ones in $[1.7, 3.3]$. If $d_0 < d_u - 2 \times 10^{-8}$, the union $L_1 \cup L_2$ is selected. If $d_0 > d_u + 2 \times 10^{-8}$, it is never selected, $d(t)$ stays above $\min(d_0, d_u + 0.005)$, and for every $\varepsilon > 5 \times 10^{-8}$ it eventually remains within ε of d_s.

Proof. All inequalities used are those certified in check (27), together with $|r| \leq 10^{-10}$. Since $F(d_u) = 0$ and $F' > 0.005$ within 0.005 of d_u, $F(d) + r$ has the sign of $d - d_u$ whenever $2 \times 10^{-8} < |d - d_u| \leq 0.005$. In the first case d therefore decreases to $d_u - 0.005$, then through $[0.12909, d_u - 0.005]$, where $F < -3.2 \times 10^{-5}$, below the coherence radius 0.1291 in finite time, and it stays below it, because $F < 0$ at 0.12909 and the only zero of F below that point is d_m. The units L_1 and L_2 are then coherent and walled from the right side, so the union is selected after the window τ_p. In the second case d increases to $d_u + 0.005$ or starts above it, the sign bands keep it in $[d_u + 0.005, 1.6]$, and on $[d_s - 0.005, d_s + 0.005]$ the slope bound $F' < -0.002$ confines it eventually to within $10^{-10}/0.002$ of d_s; so L_1 and L_2 are never coherent. □

The handoffs whose gap was recorded, from two states, have gaps between 0.59 and 0.63 at $t = 2.6$, after all four first-stage handoffs (checks (9), (10) and (14)), and so lie in the first case; all fifteen tested states unite. The shifted states of check (10) move L_1 and L_2 apart after the first stage. Gaps of 0.595, 0.795 and 0.895 unite, at $t = 7.6$, 18.4 and 61.6; gaps of 1.095 and 1.395 remain near the separated state 1.0831. Every one of these states is walled, since its gap exceeds 0.408, so all of them lie in the domain of the selector. They are produced by shifting post-handoff states, not by applying the handoff to trajectories along which the first stage has not occurred, so they are not the counterfactual class of the standard (chapter 6).

Remark 5.6 (Controls and depth (numerical)). The control without feedback unites from every shifted handoff of check (10), gaps 0.45 to 1.25, at times between 5.9 and 11.4. At synchrony its

gap equation is eq. (5.3) with $a = \lambda/(2T) = 3$, which has only the merged zero (check (9)), as proposition 3.4 predicts; the tempered label produces the same events as the control (check (8)). With record gap $s = 1.2$ or 1.3 the first stage is the same, but the handoff gap, 0.996 or 1.226, exceeds the certified middle root, 0.917 or 0.893, by more than 0.005, so by the argument of corollary 5.5, with the certified zeros of these equations, the parents never unite and remain near the separated zero, 1.198 or 1.300 (checks (14) and (27)). The hierarchy stops at depth one there, while the control unites (check (11)). The label-preserving null selects nothing for these record gaps: the proof of proposition 5.1 applies for $s \in [1.0, 1.3]$, with evaluator term at most 1.6×10^{-5} against the record term 2.5×10^{-5}. With $s = 1.0$ the gap equation has only the merged zero (check (27)), and the members of L_1 and L_2 become coherent with one another before either group is walled, so a single block of six forms at the first stage. The lower edge of the record gaps at which the gap equation is bistable is the saddle-node between 1.0722 and 1.0723: below it every post-handoff state in the ranges of proposition 5.4 unites, whatever the event history. With $s = 1.1$ the middle root is 0.963 for $n = 3$, 0.771 for $n = 4$ and 0.593 for $n = 6$, and there is no separated zero for $n = 2$ (checks (13) and (27)). For large n, replacing w_n by $\frac{n}{n-1}e^{-ad^2}$ in $F = 0$ gives

$$d_u^2 \approx \frac{2T}{n\lambda} \log \frac{2nd_u}{(n-1)\rho(s-d_u)}, \tag{5.4}$$

which reproduces 0.593 at $n = 6$. The capture distance falls with block size faster than $n^{-1/2}$ in the computed range, with local exponents between 0.65 and 0.77 from $n = 3$ to $n = 6$, because the logarithmic factor of eq. (5.4) also decreases with n: larger blocks unite only with nearer partners. This suggests that a hierarchy whose higher-level clusters are farther apart stops at finite depth under exact label sharing; we have not proved it.

Chapter 6

The formation standard

Definition 6.1 (The formation standard, restated). The definitions are those of the inheritance chapter of the main manuscript, specialized to deterministic constructions. A construction declares its objective and its flow Φ, an admitted class of initial configurations, a selector family S_c with constants c in a set C, and two rules: a parent protocol rule, which assigns to each block I a class A_I of protocols that address the block only through channels and records that read it as a whole, and a child-level probe rule, which assigns to I a class B_I of probes that agents, channels, records or declared external queriers can apply to individual members before a handoff of I, and which contains A_J for every block J merged into I. Both rules depend only on the block and on the constitutive data. A cost is a peer cost when its compared law depends on the law of another agent. The label-preserving null deletes the peer costs and keeps all labels and priors; the renormalized null also removes the labels of the deleted costs; the frozen-environment control holds every other agent's law at its initial value in each agent's force. The selector is evaluated along every control through the original formulas, and after a handoff a control remains the fine flow. The partition sequence $P(x)$ of an initial configuration x is the sequence of selected partitions, two sequences being identified when they differ by the exchange of consecutive merges of disjoint sets of agents. The clauses are these.

(F1) The admitted class is a nonempty set of initially unpartitioned configurations, open in the chart topology of the declared dynamics (full grade), or relatively open in a declared invariant family that is not contained in finitely many orbits (family grade).

(F2) The selector belongs to a family S_c, $c \in$ C, fixed before the flows are analyzed, with every real constant ranging over a set with nonempty interior and a stated persistence criterion, and each handoff is a measurable function of the state or of accessible history.

(F3) Each selected block I carries an inherited parent: a Borel retention R_I on a Borel invariant set Ω_I that contains the reachable post-handoff set $\Omega_I^{\mathrm{reach}}$, through which the parent laws (L), the dynamics (D), the joint kernels of the responses of protocols in A_I and the next retained state (B), and every channel and evaluator read (C) factor.

(F4) For each handoff some $b \in B_I$ has different response laws at two configurations of $\Omega_I^{\mathrm{reach}}$ with the same retained value.

(F5) Against each null and for each $c \in$ C, there is a nonempty open subset of the admitted class on which $P(x)$ differs from the null sequence $P^0(x)$, both taken on $[0, \infty)$; the clause holds when this is true against both nulls and for every $c \in$ C.

(F6) (F6a) The parents select a block containing at least two first-stage parents, by the same selector family, constants and constitutive data, without resetting their source data. (F6b) A counterfactual class $\mathcal{C}$ is declared beforehand. It contains the actual first-stage handoff

states and the states obtained by applying the same retentions, at the same handoff times, to trajectories from a declared class $\mathcal{U}$ containing the admitted class along which the first stage has not occurred, and it excludes states outside the domain of the selector. The second-stage event occurs after the actual handoffs and does not occur from a subset of $\mathcal{C}$ with nonempty interior.

A construction that meets (F1), (F2), (F3) and (F6a) and is shown to fail one of (F4), (F5) and (F6b) is a two-stage selection with exact inheritance. A clause that is neither established nor shown to fail leaves the classification undetermined. The strong grade of (F5) holds when (F5) holds at every admitted point, for every $c \in \mathsf{C}$, against both nulls. Two of these requirements are read here as the inheritance chapter currently states them, and the verdicts below depend on them: a verdict of (F5) that holds against one null and fails against the other fails, and (F5) must hold over a declared constant set with nonempty interior.

Definition 6.2 (Declared classes of the instance). The parent protocol rule gives each unit U the read-only class A_U. At a time fixed in advance, or chosen as a measurable function of earlier responses, a receiver outside U queries the parent A_U and receives a fresh draw from its presented law P_U; no record is assimilated by any member, and no member is named. The child-level probe rule gives B_U: a receiver attending to one member i of U through an identity channel receives a fresh draw from q_i, and a receiver querying the parent of a unit merged into U receives a fresh draw from its presented law. The selector family is that of definition 1.1 with the single constants $\delta_c = 0.05$, $\delta_w = 0.5$ and $\tau_p = 2$, and persistence on the finite window τ_p. The rules and the selector are fixed before the retention and the flows. No counterfactual class of the type of (F6b) is declared; corollary 5.5 concerns shifted post-handoff states.

Each clause is now checked for the instance of chapter 5, with its status stated as proved, numerical or open.

(F1) Proved. The admitted class eq. (5.1) is a nonempty open subset of $\mathbb{R}^{12}$, the parameter space of the declared restricted-family flow, and its states are initially unpartitioned; this is the full grade.

(F2) Proved in form, with an open requirement. The selector reads the presented laws and its own candidate clocks, so each handoff is a measurable function of accessible history; the selected blocks are nontrivial and are never dissolved, and persistence is on the window τ_p. After a handoff each difference between two members' deviations moves, by theorem 4.1(iv), along a segment from its value at the handoff to its record-determined limit, which is at most 0.0015 in absolute value in the first stage and 0.007 in the unions, far below the coherence radius 0.129. The constants are single values, so the requirement of a constant set with nonempty interior is not met as declared. Declaring such a set moves the question to (F5): the label-preserving null selects nothing for every $\delta_c \le 0.0588$ and all δ_w and τ_p, by the proof of proposition 5.1, while the shared-label flow was run only at the declared constants.

(F3) Proved at a fixed partition; open for the hybrid process. By theorem 4.1 the retention to unit means satisfies (L), (D), (C), and (B) for the read-only class A_U, between handoffs. The candidate clocks, the sizes, record means and prior rows that enter later handoffs, the merge at a union and the descent of selection events are not shown to be functions of the retained state (remark 4.2), and remark 4.4 states a route. Condition (G) is not declared.

(F4) Numerical at the first stage, open at the second. The witness probe $b \in B_U$ is a receiver attending to one member through an identity channel, which reads $q_i = \mathcal{N}(y_i, \Lambda^{-1})$ and so separates two configurations with the same unit means and different deviations. Such pairs must lie in the reachable post-handoff set, the image of $\mathcal{A}$ under the hybrid flow. At the first stage numerical constructions of admitted pairs with equal unit means after the handoffs exist, also when the four handoff times are added to the retained data; their agent means differ by about 7×10^{-5}, and these constructions are not among the checks. At the second stage the union handoff occurs a window τ_p after the left parent gap enters the coherence set, and before and after the union the retained left-side state evolves autonomously up to the weights neglected in proposition 5.4. On the reachable set the relative coordinate $\bar{y}_{L_1} - \bar{y}_{L_2}$ is then, up to those weights and to first-stage deviations, a function of the union mean and of the time since the union. A parent type that records its formation time therefore forgets, at the second stage, only first-stage deviations and terms below the resolution of the integration; a type without formation times forgets the relative coordinate. The document does not declare which, so (F4) at the second stage is open. In both stages the separation is transient and the fibers are not orbits of a symmetry of the fine field (remark 4.6).

(F5) At the declared constants, both null histories are proved and the full-flow history is numerical; over a constant set, open. By proposition 5.1 the label-preserving null never selects on $\mathcal{A}$, and by proposition 5.2 the renormalized null selects only $\mathcal{V}$. The shared-label flow first selects blocks of three (remark 5.3), so $P \neq P^0_{\mathrm{lp}}$ and $P \neq P^0_{\mathrm{rn}}$ on the fifteen tested states and, by the margins of remark 5.3, on an open neighborhood of the fixed draw, under the proviso on integration error stated there. Selector-free discrepancies exist: after the unions the gap between the member means of L_1 and L_2 relaxes by (iv) to $\rho(1+\rho)^{-1}s = 0.0055$, while under the label-preserving null it stays within 0.01 of $s = 1.1$; after the first stage the two sides stay at distance at least 3.4 by the forward invariance in proposition 5.4, while under the renormalized null every agent converges to $\rho o_i/(1+\rho)$, within 0.016 of the evaluator mean. The frozen-environment control selects the four groups and no union on thirteen tested states (check (28)), so $P \neq P^{\mathrm{fr}}$: the unions do not follow from each agent's response to the initial laws of the others. The strong grade is not established, since it needs (F5) at every admitted point; amplification is not assessed.

(F6a) Numerical. The parents A_{L_1} and A_{L_2} select their union under the inherited parent flow (iii), by the same selector with the same constants, with no reset of their source data and no new parameter, on every tested state. Only the first stage rests on integration; given the recorded handoff gaps, the unions follow from corollary 5.5.

(F6b) Proved for shifted post-handoff states; open against the class of the standard. By corollary 5.5 no union forms from post-handoff states in the stated ranges whose gap exceeds $d_u + 2 \times 10^{-8}$, a set with nonempty interior, while the actual handoffs unite. These states are not handoffs of trajectories along which the first stage has not occurred, and no class $\mathcal{U}$ of such trajectories is declared. The natural candidates point the other way. From initial states whose group spreads are those of the fixed draw scaled by 1.5, 2 or 3, the first stage has not occurred by $t = 2.3$, the last first-stage handoff time on the evaluation grid, and the four groups handed off at that time have left parent gaps 0.44, 0.30 and 0.20, all below d_u (check (28)), so by corollary 5.5 the union forms from them. The dependence of the second stage on the first that corollary 5.5 shows is a selection between two basins by the post-handoff gap; whether some declared class $\mathcal{U}$ makes it counterfactual dependence in the sense of the standard is open (open problem 8.3).

Clause	Gaussian location instance, declared selector	General laws
(F1)	proved, full grade	holds for the same class
(F2)	proved in form; single constants, so the constant set is open	as in the instance
(F3)	proved at a fixed partition, theorem 4.1; hybrid process open	universal pooled closure fails outside Gaussian families of the sufficient statistic, theorem 4.5; other retentions open
(F4)	first stage numerical; second stage open (formation times); transient, not a symmetry quotient	open
(F5)	both nulls proved on $\mathcal{A}$; full flow numerical at the declared constants; $P \neq P^{\mathrm{fr}}$ numerical; over a constant set open; strong grade not established	not established
(F6a)	numerical first stage, certified inequalities after it	not established
(F6b)	shifted post-handoff states proved, corollary 5.5; class of the standard open	not established

The classification is therefore undetermined: the construction is not shown to be a recursive formation, and the clauses that remain open are the constant set of (F2) and (F5), the hybrid part of (F3), the second stage of (F4), and (F6b) against a declared counterfactual class, with the first-stage history of the shared-label flow numerical. Further qualifications also mark where it falls short of the preferred form of the open problem on recursive formation in the inheritance chapter, which asks for a derived selector and a derived interface. The selector thresholds are declared; chapter 7 replaces them by a comparison of free energies. The label-sharing handoff is declared, and the compression interface is derived from it; the operational equivalence on which the parent is exact appears at the handoff and is absent before it, but it is created by the declared change of model, not by unchanged dynamics. The compressed coordinates are transient, and on the attracting set the members are determined by their parent and their own records. Exactness is a property of Gaussian families of the sufficient statistic (theorem 4.5). The parameter window is narrow: below the saddle-node near $s = 1.0722$ every post-handoff state unites, with record gap 1.0 the first stage already fuses the two groups, and with gaps 1.2 and 1.3 the hierarchy stops at depth one.

For general laws the selector, the shared row and its n-fold sharpening are unchanged (proposition 3.2). The pooled parent is not exactly inherited outside Gaussian families of the sufficient statistic, so (F3) is open there, (F4) depends on it, and (F5) and (F6) have not been established beyond the instance.

Chapter 7

A derived selector

The selector of definition 1.1 declares two thresholds and a window. This chapter replaces the thresholds by a comparison of free energies: a block is selected when the models that the handoff would create have a lower free energy than the current ones. The handoff rule definition 3.1, the flows and the instance are unchanged, and the window τ_p remains declared.

Definition 7.1 (Structure free energy and the derived selector). Let $\mathcal{U}$ be the current units. A coarsening of $\mathcal{U}$ is a partition $\mathcal{Q}$ of $\mathcal{V}$ each of whose blocks is a union of units of $\mathcal{U}$. For a block $U \in \mathcal{Q}$ with n_U agents, let $\mathcal{F}_U(\mathcal{Q})$ be the minimum over rows of the block free energy eq. (3.1) of U under one shared label, whose sources are the evaluator, the parent A_U when $n_U \geq 2$, and the other blocks of $\mathcal{Q}$, each presenting its law as in definition 3.1(b), with the priors of definition 3.1(a); for a single agent it is the minimum of eq. (1.1) over the row. The structure free energy of $\mathcal{Q}$ is

$$\mathcal{F}(\mathcal{Q}) = \sum_{U \in \mathcal{Q}} \mathcal{F}_U(\mathcal{Q}), \tag{7.1}$$

and $\mathcal{F}(\mathcal{U})$ is the sum of the current free energies of the units. At each time let $\mathcal{Q}^*(t)$ be a minimizer of $\mathcal{F}$ over the coarsenings of $\mathcal{U}(t)$ at the current laws. A block of $\mathcal{Q}^*(t)$ that is not a unit is a candidate, and a handoff selects it once it has been a candidate without interruption for a time τ_p. Several candidates may be selected at one time, and a selected block is never dissolved.

The record terms of eq. (7.1) are the same for every coarsening at fixed laws, so the comparison reads only the laws and the source priors. The sum runs over all blocks of $\mathcal{Q}$, because merging some units also changes the sources of the others. The selector is a measurable function of the history of the laws, through the candidate clocks, as (F2) requires; the clocks belong to the state of the process, as those of definition 1.1 do. By the readout convention of the standard (definition 6.1), the same selector is applied along every control: it evaluates eq. (7.1) with the actual costs at the laws of the control, and after a handoff along a control it evaluates the blocks' pools computed from the fine states. At $T = 1$ each $\mathcal{F}_U$ is the variational free energy of a normalized model of the members' contents and records with the laws of the other blocks as data, so minimizing eq. (7.1) compares evidence bounds under a uniform prior over coarsenings. This is structure learning in the sense of Czégel et al. [3], and greedy merging by such comparisons is the scheme of Bayesian hierarchical clustering [8].

Proposition 7.2 (The structure free energy reads spreads and pools). Under the hypotheses of definition 3.1, let every parent present the linear pool of its members. For every coarsening $\mathcal{Q}$

and every block $U \in \mathcal{Q}$,

$$\mathcal{F}_U(\mathcal{Q}) = \sum_{i \in U} D_{\mathrm{KL}}(q_i \| \bar{q}_U) - T \log\Big(\pi_{UA} + \sum_{\nu} \pi_{U\nu} e^{-n_U D_{\mathrm{KL}}(\bar{q}_U \| \nu_\nu)/T}\Big) + \sum_{i \in U} \int -\log \ell_i(o_i \mid k) q_i(dk), \tag{7.2}$$

where ν runs over the evaluator and the other blocks of $\mathcal{Q}$, and $\pi_{UA} = 0$ when $n_U = 1$. If the compared laws lie in a regular exponential family E and every parent presents the moment projection P_U of its pool, the same holds with $\sum_{i \in U} D_{\mathrm{KL}}(q_i \| P_U)$ as the first term and $D_{\mathrm{KL}}(P_U \| \nu_\nu)$ in the exponent. In the Gaussian location family of theorem 4.1, with homogeneous priors π_s and π_p and evaluator mean m_G, the structure free energy minus the record terms is

$$\frac{1}{2} \sum_{U \in \mathcal{Q}} \sum_{i \in U} \|y_i - \bar{y}_U\|_\Lambda^2 - T \sum_{U \in \mathcal{Q}} \log\Big((n_U - 1)\pi_p + \pi_s e^{-\frac{n_U}{2T}\|\bar{y}_U - m_G\|_\Lambda^2} + \sum_{V \neq U} n_V \pi_p e^{-\frac{n_U}{2T}\|\bar{y}_U - \bar{y}_V\|_\Lambda^2}\Big). \tag{7.3}$$

Proof. By eq. (3.2) the summed cost of a source ν for the members of U is $\sum_{i \in U} D_{\mathrm{KL}}(q_i \| \bar{q}_U) + n_U D_{\mathrm{KL}}(\bar{q}_U \| \nu_\nu)$, and that of the parent is the first term alone. The minimum over rows of $\sum_\nu \beta_\nu c_\nu + T D_{\mathrm{KL}}(\beta \| \pi)$ is $-T \log \sum_\nu \pi_\nu e^{-c_\nu/T}$, the Gibbs variational principle. Factoring $e^{-\sum_i D_{\mathrm{KL}}(q_i \| \bar{q}_U)/T}$ out of the sum gives eq. (7.2). For moment projections, eq. (3.4) gives $D_{\mathrm{KL}}(\bar{q}_U \| \nu_\nu) = D_{\mathrm{KL}}(\bar{q}_U \| P_U) + D_{\mathrm{KL}}(P_U \| \nu_\nu)$ for every $\nu_\nu \in \mathrm{E}$, and eq. (3.2) with P_U in place of ν gives $\sum_i D_{\mathrm{KL}}(q_i \| \bar{q}_U) + n_U D_{\mathrm{KL}}(\bar{q}_U \| P_U) = \sum_i D_{\mathrm{KL}}(q_i \| P_U)$, which is also the parent's cost. In the Gaussian location family $D_{\mathrm{KL}}(\mathcal{N}(y, \Lambda^{-1}) \| \mathcal{N}(m, \Lambda^{-1})) = \frac{1}{2}\|y - m\|_\Lambda^2$, and by definition 3.1(a) the parent of U receives the mass $(n_U - 1)\pi_p$ that the members gave to one another, a block V the mass $n_V \pi_p$, and the evaluator π_s. □

Check (18) confirms both general forms on a six-point alphabet, and check (20) confirms eq. (7.3) against explicit Gibbs rows to 6×10^{-14}. The first term of eq. (7.2) is n_U times the generalized Jensen–Shannon divergence of the members with equal weights [10]; in the Gaussian location family it is the within-block sum of squares in the Fisher metric, the objective of centroid clustering [11]. The second term is n_U times the free energy, at temperature T/n_U, of a single receiver that holds the pool and has one additional source at cost zero, the block's own parent, carrying the mass that the members gave to one another. For a block far from every other source the second term is close to $-T \log((n_U - 1)\pi_p)$, and eq. (7.3) becomes a centroid criterion with a penalty for each block, of the kind that Dirichlet-process mixtures give in the limit of small variance [9]; here the penalty depends on the block size and is fixed by the source prior instead of being declared.

Proposition 7.3 (Mutual attention favors a merge). Let $i \neq j$ be single-agent units that compare each other through identity channels, with Gibbs rows β_i, β_j, mutual priors π_{ij}, π_{ji} and costs D_{ij}, D_{ji}, and let $\mathcal{Q}$ merge them, so that their parent has prior $\pi_A = (\pi_{ij} + \pi_{ji})/2$ by definition 3.1(a). Let the parent present the linear pool, or the moment projection onto a family that contains q_i and q_j. Then

$$\mathcal{F}_i + \mathcal{F}_j - \mathcal{F}_{\{i,j\}}(\mathcal{Q}) \geq T \log \frac{\beta_{ij}\beta_{ji}\pi_A}{\pi_{ij}\pi_{ji}} + \max(D_{ij}, D_{ji}). \tag{7.4}$$

Let an agent v outside the pair compare i and j through one Markov channel, with costs D_{vi}, D_{vj} and current attention $\beta_{v\{ij\}} = \beta_{vi} + \beta_{vj}$ to the pair, and let it attend to the parent through the same channel with prior $\pi_{vi} + \pi_{vj}$. If the parent is the linear pool, or in the Gaussian location family its

moment projection, then

$$\mathcal{F}_v(\mathcal{Q}) - \mathcal{F}_v \le -T \log\Big(1 - \beta_{v\{ij\}}\big(1 - e^{-|D_{vi} - D_{vj}|/(2T)}\big)\Big). \tag{7.5}$$

Proof. Write Z_i for the normalizer of the row of i. Since $\beta_{ij} = \pi_{ij} e^{-D_{ij}/T}/Z_i$, we have $\mathcal{F}_i = -T \log Z_i = -T \log \pi_{ij} + D_{ij} + T \log \beta_{ij}$, and likewise for j. The normalizer of the pair's shared row contains the term of its parent, so $\mathcal{F}_{\{i,j\}}(\mathcal{Q}) \le -T \log \pi_A + D_{iA} + D_{jA}$, where D_{lA} is the cost of member l for the parent. In both cases the parent's law minimizes the summed cost $D_{\mathrm{KL}}(q_i \| p) + D_{\mathrm{KL}}(q_j \| p)$ over the laws p it may present, by eq. (3.2) and, for the projection, eq. (3.4). The laws q_j and q_i are among them, so $D_{iA} + D_{jA} \le \min(D_{ij}, D_{ji})$. Adding the three relations gives eq. (7.4), because $D_{ij} + D_{ji} - \min(D_{ij}, D_{ji}) = \max(D_{ij}, D_{ji})$.

For v, only the terms of i and j in its normalizer Z_v change. Before the merge they sum to $b \le (\pi_{vi} + \pi_{vj}) e^{-\min(D_{vi}, D_{vj})/T}$. After it the parent's term is $(\pi_{vi} + \pi_{vj}) e^{-D_{vA}/T}$, and $D_{vA} \le \frac{1}{2}(D_{vi} + D_{vj})$: for the pool by convexity of D_{KL} in its second argument, after the channel, which maps the pool to the pool of the channelled laws; in the Gaussian location family by convexity of the squared norm. Since $\frac{1}{2}(D_{vi} + D_{vj}) - \min(D_{vi}, D_{vj}) = \frac{1}{2}|D_{vi} - D_{vj}|$, the new term is at least $b e^{-|D_{vi} - D_{vj}|/(2T)}$, so the new normalizer is at least $Z_v - b\big(1 - e^{-|D_{vi} - D_{vj}|/(2T)}\big)$, and $b = \beta_{v\{ij\}} Z_v$. □

The first term of eq. (7.4) is a saving of labels, the Occam factor of Bayesian model comparison [12]: two agents that attend to each other pay for two independent choices of source in their own models and for one in the shared model, whose parent collects the mass they gave to each other. The second term is a double use of the members' laws. The parent presents the law closest to both members in summed divergence, so each member is compared with a law that contains itself, and the bound grows with the divergence between the members as long as they attend to each other. A cavity parent, which presents to each member the pool of the other members, removes the double use: for a pair its cost is $D_{ij} + D_{ji}$, and the bound keeps only its first term. It also makes the cost of the own parent differ from the cost common to all sources by a multiple of the spread, so the shared row depends on the spread and theorem 4.1 fails. The saving of labels remains with the cavity parent. Check (19) confirms both bounds on 2000 random draws on a five-point alphabet, with smallest slacks 0.088 and 0.0011.

Corollary 7.4 (The derived selector reads initial attention). Let the peer priors be homogeneous, $\pi_{ij} = \pi_p$. At a state where two agents satisfy

$$T \log \frac{\beta_{ij}\beta_{ji}}{\pi_p} + \max(D_{ij}, D_{ji}) > -T \sum_{v \ne i,j} \log\Big(1 - \beta_{v\{ij\}}\big(1 - e^{-|D_{vi} - D_{vj}|/(2T)}\big)\Big), \tag{7.6}$$

the partition into single agents does not minimize eq. (7.1), so the derived selector has a candidate there. In particular, if such a pair exists at a state of the admitted class, the selector has a candidate at time zero along both Φ and Φ^0.

Proof. By proposition 7.3, merging the pair lowers $\mathcal{F}$ by at least the left side minus the right side of eq. (7.6). The two flows start from the same state, and the selector is the same function of the state. □

With the declared selector an admitted class could be chosen on which the label-preserving null flow never selects (proposition 5.1), because the coherence radius 0.129 lies below the spacing of the records. The derived selector has no radius to choose. It groups agents that attend to one another, and the records alone make agents attend to one another when nothing else lies closer. The criterion detects attention, not agreement: by eq. (7.4) the members' gain grows with $\max(D_{ij}, D_{ji})$ as long as they attend to each other, so the selector merges two single agents that attend mainly to each other at any divergence, and a merge radius exists only between blocks, whose own-parent mass caps the attention term (proposition 7.6). The comparison is therefore derived relative to a declared model family, a declared prior over coarsenings and a declared window. A first stage is interaction-necessary only if the interaction changes the minimizer of eq. (7.1) within the window. At the fixed draw of the instance, $\beta_{01}\beta_{10} = 0.254$ and $\beta_{12}\beta_{21} = 0.229$ exceed $\pi_p = 0.086$; for the pair $\{0, 1\}$ the members gain 1.737 against the bound 1.145, and the outsiders' free energies change by -0.0035. Over all $4{,}213{,}597$ partitions of the twelve agents, the minimizer of eq. (7.1) at the fixed draw is the four groups, with margin 1.715 to the next (check (21)). The enumeration can be reduced exactly. Every attention term has an argument at most $\pi_s + (N-1)\pi_p = 1$, so the structure free energy is at least $\frac{\lambda}{2}$ times the within-block sum of squares. A block that holds a left and a right agent, at distance at least 3.59 at the fixed draw, contributes at least $\frac{\lambda}{2} \cdot \frac{1}{2}(3.59)^2 > 19$, more than the value 7.563 of the four groups, so the minimizer respects the sides, and up to cross-side weights below e^{-38} it is the union of the minimizers over the 203 partitions of each side.

Remark 7.5 (Derived selections in the instance (numerical)). Take the instance of chapter 5 with the derived selector in place of definition 1.1. The minimizer was computed over the interval coarsenings of the units ordered by their means. In one dimension interval coarsenings need not contain the minimizer over all coarsenings, since the attention terms reward close means and can favor interleaved blocks, but at the fixed draw the interval minimizer agrees with the minimizer over all partitions, and at five later states with the minimizer over the $41{,}209$ partitions that do not mix the two sides (checks (21) and (22)); after the first stage, with four units, every coarsening that is not an interval coarsening mixes the sides and is excluded by the spread bound in the proof of proposition 7.6. Along the flow with individual rows from the fixed draw, the minimizer is the four groups from $t = 0$, with margin at least 0.86 on $[0, 1.3]$. From $t = 1.75$ the minimizer is $\{L_1, L_2, R_1 \cup R_2\}$, and from $t = 1.77$ it is $\{L_1 \cup L_2, R_1 \cup R_2\}$ (check (22)). The gap between the means of L_1 and L_2 falls below $d_u - 0.005$ at $t = 0.96$. On the box $\prod_i [o_i - 0.005, o_i + 0.005]$, which is forward invariant under the label-preserving null by the proof of proposition 5.1, also after handoffs because the control then remains the fine flow, the interval minimizer is the four groups at each of 200 sampled states, with margin at least 1.82 among interval coarsenings; over all coarsenings the margin at the fixed draw is 1.715 (check (21)). The selector is evaluated after every integration step of length 0.05, so a candidate present from the start is selected at $\tau_p + 0.05$ on that grid, and at τ_p under the definition.

Three regimes follow (check (23)). With $\tau_p = 0.5$ the shared-label flow and the label-preserving null select the four groups at $t = 0.55$, before the left gap has fallen below $d_u - 0.005$; the shared-label parents stay separated, with left gap 1.056 at $t = 300$, and the two partition sequences coincide. With $\tau_p = 1.25$ both select the four groups at $t = 1.30$, the shared-label flow then selects $L_1 \cup L_2$ and $R_1 \cup R_2$ at $t = 40.3$ and 44.15, and the null selects nothing more; over twelve further states of $\mathcal{A}$ the partition sequences are the same, with unions between $t = 36.5$ and 60.9 (check (24)). With

$\tau_p = 2$, the window of chapter 5, the minimizer changes before the window ends, the shared-label flow selects $L_1 \cup L_2$ and $R_1 \cup R_2$ as first-stage blocks at $t = 3.75$ and 3.80, the label-preserving null selects the four groups at $t = 2.05$, and there is no second stage. At each of the three windows the renormalized null selects the two sides of six and then all twelve agents, and the frozen-environment control selects the four groups and nothing more (check (28), fixed draw). The switch times of the fixed draw suggest that two stages form whenever the handoff falls in $[0.96, 1.75)$, that is for $0.96 \le \tau_p < 1.75$ under the definition and $0.91 \le \tau_p < 1.70$ on the evaluation grid; only $\tau_p = 1.25$ was run in that range. A handoff in $[1.75, 1.77)$, while the minimizer is $\{L_1, L_2, R_1 \cup R_2\}$, gives a mixed regime in which the right side selects $R_1 \cup R_2$ at the first stage; it is not among the checks. The control without feedback, with $\tau_p = 1.25$, selects the four groups at $t = 1.30$ and their unions at $t = 3.0$ and 3.05.

Proposition 7.6 (The derived union criterion is a radius). After the four first-stage handoffs, let the parent means lie in the ranges of proposition 5.4 and write $d = m_2 - m_1$. The minimizer of eq. (7.1) over the coarsenings of the four units contains $L_1 \cup L_2$ when $\Gamma_3(d) > 10^{-10}$ and does not when $\Gamma_3(d) < -10^{-10}$, where, for two parents of n members each,

$$\Gamma_n(d) = \log \frac{2n-1}{\pi_p} - 2\log\big(n - 1 + ne^{-n\lambda d^2/(2T)}\big) - \frac{n\lambda}{4T} d^2. \tag{7.7}$$

If $(2n-1)\pi_p < 1$, then Γ_n has exactly one zero $d_*(n)$ in $(0, \infty)$, is positive below it and negative above it, and is decreasing above it. In the instance, with $T = 1$ and $\lambda = 6$, $d_*(3) = 0.7685$.

Proof. For the union, $\sum_{i \in L_1 \cup L_2} (y_i - \bar{y})^2 = W_{L_1} + W_{L_2} + \frac{n_1 n_2}{n_1 + n_2} d^2$, the decomposition of a sum of squares about the pooled mean, so the spreads of L_1 and L_2 cancel between the two coarsenings, whatever their values, and the union adds $\frac{\lambda}{2} \cdot \frac{3}{2} d^2$. By eq. (7.3) each separate parent contributes $-\log(2\pi_p + 3\pi_p e^{-9d^2})$ and the union $-\log(5\pi_p)$, once the evaluator and the right units are dropped from the normalizers. As in the proof of proposition 5.4, the evaluator's weight relative to $2\pi_p$ is below 1.5×10^{-12} for a left parent and smaller still for the union, and every weight between the two sides is below e^{-100}, which also bounds the change in the right units' free energies. Hence $\mathcal{F}(\text{separate}) - \mathcal{F}(\text{union}) = \Gamma_3(d) + e$ with $|e| < 10^{-10}$. Every other coarsening joins a left and a right unit, whose means differ by at least 3.4, so its spread term is at least $\frac{\lambda}{2} \cdot \frac{3}{2}(3.4)^2 > 50$; each attention term of a block of at least three members lies in $[0, -\log(2\pi_p)]$, so the attention terms of two coarsenings of the four units differ by less than 8, and such a coarsening has a larger structure free energy than the four separate units. The choices on the two sides are independent up to the same weights.

For the zeros put $u = d^2$ and $a = n\lambda/(2T)$. The function $h(u) = \log(n - 1 + ne^{-au})$ has $h''(u) = a^2 n(n-1)e^{-au}/(n-1+ne^{-au})^2 > 0$, so Γ_n is strictly concave in u. It has the value $-\log((2n-1)\pi_p) > 0$ at $u = 0$ and tends to $-\infty$. A strictly concave function with these properties has exactly one zero, is positive before it, negative after it, and decreasing after it, because its maximum lies before the zero. □

The radius lies between the merged root and the middle root of the gap equation, $d_m = 0.00456 < d_* = 0.7685 < d_u - 0.005 = 0.958$. Check (27) certifies $d_*(3) \in (0.76849, 0.76851)$, $\Gamma_3 < -1.5$ on $[0.96804, 1.6]$ and $\Gamma_3 < -2.5$ on $[1.09, 1.11]$. For other block sizes $d_*(2) = 1.087$, $d_*(4) = 0.600$ and $d_*(6) = 0.410$; the value $\Gamma_n(0) = -\log((2n-1)\pi_p)$, the saving of labels when two synchronized

parents merge, shrinks as the parents' share of the prior grows, down to $-\log(1-\pi_s) = 0.051$ when the union holds all agents. Check (25) confirms the reduction to within 2×10^{-12} over 72 configurations in the stated ranges.

Corollary 7.7 (Shifted post-handoff states under the derived selector). Consider post-handoff states as in corollary 5.5, with gap d_0 and no running candidate clock. If $d_0 < d_u - 2 \times 10^{-8}$, the union $L_1 \cup L_2$ is selected. If $d_0 > d_u + 2 \times 10^{-8}$, the gap stays above $\min(d_0, d_u + 0.005)$ for all time and the union is never selected.

Proof. By the proof of corollary 5.5, with the certified inequalities of check (27), in the first case d falls below d_*, which lies in $[0.12909, d_u - 0.005]$, in finite time and stays below it, so by proposition 7.6 the union is a block of the minimizer from then on and is selected after τ_p. In the second case $d > d_u > d_*$ for all time, so $\Gamma_3(d) < 0$, by the monotonicity in proposition 7.6, and the union is never a candidate. □

Check (26) shifts L_1 and L_2 apart after the handoffs at $\tau_p = 1.25$. Handoff gaps of 0.59, 0.79, 0.89 and 0.94 unite, at $t = 2.65$, 5.3, 44.15 and 155.6; gaps of 0.99, 1.09 and 1.39 do not unite by $t = 2000$ and end between 1.0829 and 1.0833, near d_s. The control without feedback unites from every tested gap, 0.879 to 1.679, at times between 3.05 and 62.0.

Remark 7.8 (The formation standard under the derived selector). For handoffs in $[0.96, 1.75)$, tested at $\tau_p = 1.25$, the clauses read as follows. (F1) is unchanged. (F2) holds in form with a derived comparison: the window τ_p, the uniform prior over coarsenings, the pool parent, the reassignment of mutual prior mass to the parent and the handoff rule are declared, ties are resolved by the rule of remark 1.3, and the single value of τ_p leaves the constant set open as in chapter 6. (F3) and (F4) are as in chapter 6, open in the same respects, because the handoff rule is unchanged and theorem 4.1 holds between handoffs after any sequence of shared-label handoffs; for either selector a complete retained parent must also carry the candidate clocks. For (F5), the label-preserving null history on the tested states is the four groups at $\tau_p + 0.05$ and nothing after: on the invariant box the left gap stays within 0.01 of 1.1, so $\Gamma_3 < -2.5$ there by proposition 7.6, and likewise on the right. The selection of the four groups along this null rests on the sampled box states, and the absence of later events is proved. The renormalized null selects the two sides and then all agents, and the frozen-environment control the four groups alone (check (28), fixed draw). The shared-label flow selects the four groups and then the unions, so P differs from both null sequences and from P^{fr} on the tested states and, by the margins of remark 7.5, on an open neighborhood of the fixed draw, under the same proviso on integration error as in remark 5.3; against the renormalized null only the fixed draw was run. The first stage alone does not distinguish Φ from the label-preserving null: that null selects the same blocks at the same time, and they minimize eq. (7.1) at the initial state. The strong grade is not established. (F6a) holds on the tested states. (F6b) holds for shifted post-handoff states by corollary 7.7 and is open against the class of the standard, as for the declared selector. With $\tau_p = 2$ the flow selects the two unions as its only stage, and P differs from both null sequences on the fixed draw; with (F3) open it is not shown to be a single-stage selection with exact inheritance. With $\tau_p = 0.5$ the shared-label flow and the label-preserving null have the same partition sequence on the fixed draw, so (F5) fails at that constant.

Clause	Declared selector	Derived, handoff in $[0.96, 1.75)$	Derived, $\tau_p = 2$
(F1)	proved	proved	proved
(F2)	in form; declared thresholds and window; constant set open	in form; derived comparison, declared window; constant set open	in form; derived comparison, declared window; constant set open
(F3)	fixed partition proved, theorem 4.1; hybrid open	fixed partition proved; hybrid open	fixed partition proved; hybrid open
(F4)	first stage numerical; second stage open	first stage numerical; second stage open	numerical
(F5)	nulls proved; full flow numerical	label-preserving null partly sampled; full flow numerical	numerical, fixed draw
(F6a)	numerical	numerical	fails: no second stage
(F6b)	shifted states proved; class of the standard open	shifted states proved; class of the standard open	not applicable

The derived selector removes the two thresholds but not the window, which takes over their role. Two times matter: the time at which the interaction has brought the groups within the middle root of the gap equation, and the later time at which the minimizer itself moves away from the four groups. A first handoff before the first time stops the hierarchy at depth one, a handoff between them gives two stages, and a window that outlasts the second gives a mixed regime or a single coarser stage. Interaction necessity stage by stage is not a clause of the standard, and this instance does not give it at both stages on either side: until $t = 1.75$ the minimizer along both the shared-label flow and the label-preserving null is the four groups, so a first stage completed by then coincides with that null's; a handoff in $[1.75, 1.77)$ changes the first stage of the right side only; and from $t = 1.77$ the minimizer of the shared-label flow is the pair of unions. A prior over coarsenings that offsets the saving of labels, or a cavity parent at the cost of exact inheritance, would change the radius and the regimes (open problem 8.6).

Chapter 8

Prior work and open problems

The ingredients have precedents. Inferred group membership that feeds back on individual updating appears in the meta-contrast model of Salzarulo [19], where categories are recomputed from the opinion context and agents move toward prototypes. It also appears in the latent-group model of Gershman et al. [7], where an observer infers group structure with shared choice parameters and the posterior determines social influence. Replacing locally coherent groups by representatives that act at the next stage is the hierarchical consensus scheme of Perony et al. [18], with prescribed stages and no feedback to the members. Martins [14] derived bounded confidence from Bayesian updating, with stubbornness from shrinking individual uncertainty. Normalized state-dependent weights are the setting of Motsch and Tadmor [16], and anchoring of opinions to fixed data is the model of Friedkin and Johnsen [5]. Exact block-mean dynamics of linear consensus over an external equitable partition were analyzed by Martini et al. [13] and Egerstedt et al. [4]; O'Clery et al. [17] related them to observability and coarse graining, and Schaub et al. [20] used these partitions for cluster synchronization in nonlinear networks. Multilevel selection has been written as Bayesian inference, with the formation of new levels as structure learning [3]. We found no work that combines a shared source label for dynamically formed groups, exact block-mean closure for state-dependent rows away from synchrony, and merging that depends on the state at formation. The comparison covered the works cited here, each checked against its publisher record, and does not establish priority.

Open problem 8.1 (General laws). Find an exactly inherited compressive parent for shared-label blocks outside Gaussian families of the sufficient statistic. Candidates are a retention other than the pool, such as the canonical parent of the inheritance chapter for the routed interface, a pooled parent on a declared invariant submanifold of an exponential family, or an approximate inheritance whose defect is controlled by the spread inside the block. Alternatively, show that no exact compressive parent exists for Bernoulli blocks with Gibbs rows.

Open problem 8.2 (Hybrid inheritance). Carry out the route of remark 4.4: define the parent type with its mass, record mean, prior row and running clocks, prove that the hybrid process descends between events, at events and after composition, with the read-only class of definition 6.2, and decide whether the type records formation times, which settles the second stage of (F4). With a hard selector each handoff triggered by a transversal entry places the handoff states on a hypersurface, so a parent that records its formation time recovers one forgotten coordinate per event; a general statement of this constraint is open.

Open problem 8.3 (Counterfactual class). Declare a class $\mathcal{U}$ of initial configurations and the resulting counterfactual class of the standard for the instance, and decide (F6b) against it. The widened initial states of check (28) give handoff gaps below the middle root, from which the union

forms, so a class on which the union fails on a set with nonempty interior must produce larger post-handoff gaps without a first stage.

Open problem 8.4 (A corrected convex statement). In the Gaussian tangent global joint of Items 2 and 3 with its own flow, decide whether parents anchored at the record means of their own clusters exclude (F6b): whether the parent gap from every handoff of a declared class approaches its limit without entering the coherence set unless the limit lies in it. Decide also whether instances meeting (F6b) exist when the parents interact only through inherited bridges, and whether declared parent anchors count as new constitutive data under (F6a).

Open problem 8.5 (A derived persistence criterion). The derived selector of chapter 7 keeps the window τ_p, and the window decides whether the hierarchy stops at depth one, forms two stages, or forms one coarser stage (remark 7.8). Replace it by a criterion read from the agents' models, for example accumulated evidence for the shared label in the sense of a sequential test [21], or the message counts of the registered-agreement model of the dynamics chapter, and decide whether the two-stage regime survives.

Open problem 8.6 (Interaction necessity at both stages). Under the derived selector the first stage reads the attention present in the initial state (proposition 7.3 and corollary 7.4). Decide whether a prior over coarsenings that offsets the saving of labels, or a cavity parent together with an exact replacement for theorem 4.1, gives an admitted class on which the label-preserving null does not select the first-stage blocks while the second stage still depends on the first, and whether some instance gives interaction necessity at both stages without either change.

Open problem 8.7 (Persistent compression). The coordinates discarded here are transient. Combine the shared label with the code-invisible sectors of Items 1 to 3, or with the flatness condition of remark 3.5, so that the parent forgets coordinates that persist and that an admitted child-level probe still reads.

Open problem 8.8 (The reciprocal potential). In the per-agent model a source feels no force from the receivers that attend to it. Under the reciprocal potential a parent that presents a pool receives source-role forces from every receiver. Item 2 shows that, with means, rows and the other costs frozen, the stationary variance of a single sender with one outgoing edge increases with its disagreement; that is not a statement about pooled sources along the joint flow. Determine whether source-role widening offsets the n-fold sharpening of the shared row, and whether exact inheritance survives the source-role terms. This is the question about certainty raised in Item 3.

Open problem 8.9 (Depth). By eq. (5.4) the capture distance falls with block size, faster than $n^{-1/2}$ in the computed range. Determine the depth reached from random records, and whether partial pooling, with the members' costs added with weight n^{p-1} for some $0 < p < 1$, allows any finite depth while keeping counterfactual dependence at every stage.

Open problem 8.10 (Validated numerics). Certify the first-stage events on an explicit open subset of $\mathcal{A}$ by interval integration, on $[0, 2.6]$ for the declared selector and on $[0, 1.30]$ for the derived selector. The unions then follow from the certified inequalities of check (27), given handoff gaps below $d_u - 0.005$ (and above d_* for the derived selector), and with them the open-set form of (F5) and (F6a) at the declared constants.

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